

HOUSE PRICE PREDICTION

Problem Statement:

A house value is simply more than location and square footage. Like the features that make up a person, an educated party would want to know all aspects that give a house its value. For example, you want to sell a house and you don't know the price which you may expect — it can't be too low or too high. To find house price you usually try to find similar properties in your neighbourhood and based on gathered data you will try to assess your house price.

Problem Definition:

Real estate transactions, whether buying or selling, often present challenges for individuals and businesses. Determining an appropriate market value for a property can be complex and involves considering numerous factors. Offering a price that is either too low or too high can significantly impact the success of the transaction.

This project aims to develop a predictive model that accurately estimates the market price of a residential property based on its attributes. By analysing historical property data, we seek to understand the influence of various factors on property prices.

Key Objectives:

- Develop a predictive model: Create a machine learning model capable of accurately estimating property prices.
- Identify key factors: Determine the most influential attributes that impact property prices.
- Provide valuable insights: Gain a deeper understanding of the real estate market dynamics and how different factors contribute to property values.

Significance:

Accurate property pricing is crucial for both buyers and sellers. This model will empower stakeholders to make informed decisions, optimize their investment strategies, and achieve successful real estate transactions.

Understanding Business/Social Opportunity

Social Opportunity:

- **Empowering Homeowners:** By providing accessible and affordable guidance, we can empower individuals to make informed decisions about their property transactions.
- **Reducing Financial Risk:** Our services can help homeowners avoid potential financial losses due to mispricing, ensuring they receive fair market value for their property.
- **Promoting Fair Market Practices:** By providing accurate and transparent information, we can contribute to a more equitable and transparent real estate market.

Business Opportunity:

- **Addressing a Market Need:** There is a significant demand for expert guidance in the real estate market, especially among individuals who lack the knowledge and expertise to navigate the complexities of property transactions.
- **Revenue Generation:** By charging fees for our services, we can generate revenue and build a profitable business.
- **Building a Sustainable Business:** By providing valuable services and building strong client relationships, we can establish a sustainable and successful business in the real estate sector.

Import the Libraries which is required for Data Processing and for creating model.

Numpy Version: 1.26.4

Pandas Version: 2.2.2

Seaborn Version: 0.13.2

Matplotlib Version: 3.10.0

2 - Data Report:

Data Understanding

After loading data into our pandas library dataframe, we can now try to understand the kind of data we have with us.

Shape of Data - We have 21613 rows and 23 columns in dataset.

```
(21613, 23)
```

Columns Details –

```
Index(['cid', 'dayhours', 'price', 'room_bed', 'room_bath', 'living_measure',  
      'lot_measure', 'ceil', 'coast', 'sight', 'condition', 'quality',  
      'ceil_measure', 'basement', 'yr_built', 'yr_renovated', 'zipcode',  
      'lat', 'long', 'living_measure15', 'lot_measure15', 'furnished',  
      'total_area'],  
      dtype='object')
```

Given above columns details mentioned with their defination.

1. cid: a notation for a house. we Will drop this column.
2. dayhours: Date house was sold. we will convert into date and time.
3. price: The target feature that we aim to predict based on the other features.
4. room_bed: Indicates the number of bedrooms in the house.
5. room_bath: Indicates the number of bathrooms in the house.
6. living_measure: Represents the square footage of the house's living area.
7. lot_measure: Represents the square footage of the lot.
8. ceil: Indicates the number of floors in the house.
9. coast: Indicates whether the house has a waterfront view. It looks a categorical variable, which will be explored further in data analysis.
10. sight: Represents the number of times the house has been viewed.
11. condition: Indicates the overall condition of the house, similar to a rating assigned to it.
12. quality: Represents the grade assigned to the house based on a grading system.
13. ceil_measure: Represents the square footage of the house, excluding the basement.
14. basement: Represents the square footage of the basement.
15. yr_built: Indicates the year the house was originally built.
16. yr_renovated: Indicates the year the house was most recently renovated.
17. zipcode: Represents the zip code of the house's location, as the name implies.

18. lat: It is representing the Latitude coordinates.

19. long: It is representing the Longitude coordinates.

20. living_measure15: Represents the square footage of the house as measured in 2015. This value reflects the house area, which may or may not have changed due to renovations, if any were carried out after that year.

21. lot_measure15: lot_measure15: Represents the square footage of the lot as measured in 2015. This value reflects the lot area, which is unlikely to change but may vary if any modifications were made after that year.

22. furnished: It is showing house is furnished or not. It seems to be categorical variable as per info.

23. total_area: Measure of both living and lot

Check the Data Types of the columns for the Dataset -

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21613 entries, 0 to 21612
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   cid                   21613 non-null  int64
1   dayhours              21613 non-null  object
2   price                 21613 non-null  int64
3   room_bed              21505 non-null  float64
4   room_bath             21505 non-null  float64
5   living_measure        21596 non-null  float64
6   lot_measure           21571 non-null  float64
7   ceil                  21571 non-null  object
8   coast                 21612 non-null  object
9   sight                 21556 non-null  float64
10  condition             21556 non-null  object
11  quality               21612 non-null  float64
12  ceil_measure          21612 non-null  float64
13  basement              21612 non-null  float64
14  yr_built              21612 non-null  object
15  yr_renovated          21613 non-null  int64
16  zipcode               21613 non-null  int64
17  lat                   21613 non-null  float64
18  long                  21613 non-null  object
19  living_measure15      21447 non-null  float64
20  lot_measure15         21584 non-null  float64
21  furnished             21584 non-null  float64
22  total_area            21584 non-null  object
dtypes: float64(12), int64(4), object(7)
memory usage: 3.8+ MB
```

We have below features in our dataset.

- 4 features are of integer type
- 12 features are of float type
- 7 feature is of object type

Check the Statistical Summary (Descriptive details) of Data –

	count	mean	std	min	25%	50%	75%	max
cid	21613.0	4.580302e+09	2.876566e+09	1.000102e+06	2.123049e+09	3.904930e+09	7.308900e+09	9.900000e+09
price	21613.0	5.401822e+05	3.673622e+05	7.500000e+04	3.219500e+05	4.500000e+05	6.450000e+05	7.700000e+06
room_bed	21505.0	3.371355e+00	9.302886e-01	0.000000e+00	3.000000e+00	3.000000e+00	4.000000e+00	3.300000e+01
room_bath	21505.0	2.115171e+00	7.702481e-01	0.000000e+00	1.750000e+00	2.250000e+00	2.500000e+00	8.000000e+00
living_measure	21596.0	2.079861e+03	9.184961e+02	2.900000e+02	1.429250e+03	1.910000e+03	2.550000e+03	1.354000e+04
lot_measure	21571.0	1.510458e+04	4.142362e+04	5.200000e+02	5.040000e+03	7.618000e+03	1.068450e+04	1.651359e+06
sight	21556.0	2.343663e-01	7.664376e-01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	4.000000e+00
quality	21612.0	7.656857e+00	1.175484e+00	1.000000e+00	7.000000e+00	7.000000e+00	8.000000e+00	1.300000e+01
ceil_measure	21612.0	1.788367e+03	8.281025e+02	2.900000e+02	1.190000e+03	1.560000e+03	2.210000e+03	9.410000e+03
basement	21612.0	2.915225e+02	4.425808e+02	0.000000e+00	0.000000e+00	0.000000e+00	5.600000e+02	4.820000e+03
yr_renovated	21613.0	8.440226e+01	4.016792e+02	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	2.015000e+03
zipcode	21613.0	9.807794e+04	5.350503e+01	9.800100e+04	9.803300e+04	9.806500e+04	9.811800e+04	9.819900e+04
lat	21613.0	4.756005e+01	1.385637e-01	4.715590e+01	4.747100e+01	4.757180e+01	4.767800e+01	4.777760e+01
living_measure15	21447.0	1.987066e+03	6.855196e+02	3.990000e+02	1.490000e+03	1.840000e+03	2.360000e+03	6.210000e+03
lot_measure15	21584.0	1.276654e+04	2.728699e+04	6.510000e+02	5.100000e+03	7.620000e+03	1.008700e+04	8.712000e+05
furnished	21584.0	1.967198e-01	3.975279e-01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00

Feature Definitions and Analysis Observations -

1 - Price: Target variable representing the sale price of the property.

Observations: Mean price is 540,182, with a large spread (std: 367,362), indicating significant variability in house prices.

Minimum price is 75,000, while the maximum price is 7.7M, suggesting the presence of luxury properties or outliers.

Median price (\$450,000) is lower than the mean, suggesting a right-skewed distribution (a few very high prices).

2 - Number of bedrooms in the house.

Observations: Median number of bedrooms is 3, and most houses have between 3 and 4 bedrooms (25th–75th percentile).

There are extreme cases like a house with 33 bedrooms, likely an outlier or error.

3 - room_bath: Number of bathrooms in the house.

Observations: Median is 2 bathrooms, with a maximum of 8 bathrooms.

Bathrooms also show slight variability, but outliers are fewer compared to bedrooms.

4 - living_measure: Square footage of the living area.

Observations: Median is 1,910 sq. ft., with a maximum of 13,540 sq. ft.

Large std (918 sq. ft.) and high max value suggest significant size variability among properties.

5 - lot_measure: Square footage of the lot.

Observations: Mean is 15,106 sq. ft., but std is very high (41,420 sq. ft.), indicating outliers.

Max value of 1.65M sq. ft. (about 37 acres) suggests some lots are exceptionally large.

6 - ceil: Number of floors in the house.

Observations:

Most houses have 1 or 2 floors, with a maximum of 3 floors.

Distribution is fairly consistent, with no extreme outliers.

7 - coast: Indicates if the house has a waterfront view (1 for yes, 0 for no).

Observations: Only about 0.7% of houses have waterfront views, making this a rare feature.

8 - sight: Represents the number of views received by the house.

Observations: Most houses have 0 views, and the maximum is 4 views.

Could be used as a proxy for popularity.

9 - condition: Represents the overall condition of the house (rating 1–5).

Observations: Median condition is 3, and most houses are rated between 3 and 4. Very few houses are rated at the extremes (1 or 5).

10 - quality: Represents the grade based on the house grading system (rating 1–13).

Observations: Median is 7, indicating average or slightly above-average quality. Some houses have an exceptionally high grade of 13.

11 - yr_built and yr_renovated: Construction year and renovation year.

Observations: Most houses were built around 1971, with renovation often not performed (75% of yr_renovated is 0).

12 - furnished: Indicates whether the house is furnished (1 for yes, 0 for no).

Observations: Only about 19.6% of houses are furnished, making it a rare feature.

13 - living_measure15 and lot_measure15: Square footage measured in 2015 (living area and lot).

Observations: These values are generally consistent with living_measure and lot_measure, but smaller max values suggest updates or adjustments in 2015.

14 - Total_area: Represents the total area (living and lot).

Observations: Median is 9,575 square_feet, and max is 1.65M square_feet, reinforcing the presence of very large properties.

General Observations

1 - **Data Skewness**: Many features, including price, living_measure, and lot_measure, show right-skewed distributions due to a few very high values.

2 - **Outliers**: Features like room_bed, lot_measure, and price have extreme outliers that may need to be handled for robust modeling.

3 - **Categorical Features**: Features like coast and furnished are highly imbalanced and may require careful treatment during analysis.

4 - **Consistency**: Features like living_measure15 and lot_measure15 align with their main counterparts, suggesting consistency in measurements.

Before Renaming the column –

```
Index(['cid', 'dayhours', 'price', 'room_bed', 'room_bath', 'living_measure',  
      'lot_measure', 'ceil', 'coast', 'sight', 'condition', 'quality',  
      'ceil_measure', 'basement', 'yr_built', 'yr_renovated', 'zipcode',  
      'lat', 'long', 'living_measure15', 'lot_measure15', 'furnished',  
      'total_area'],  
      dtype='object')
```

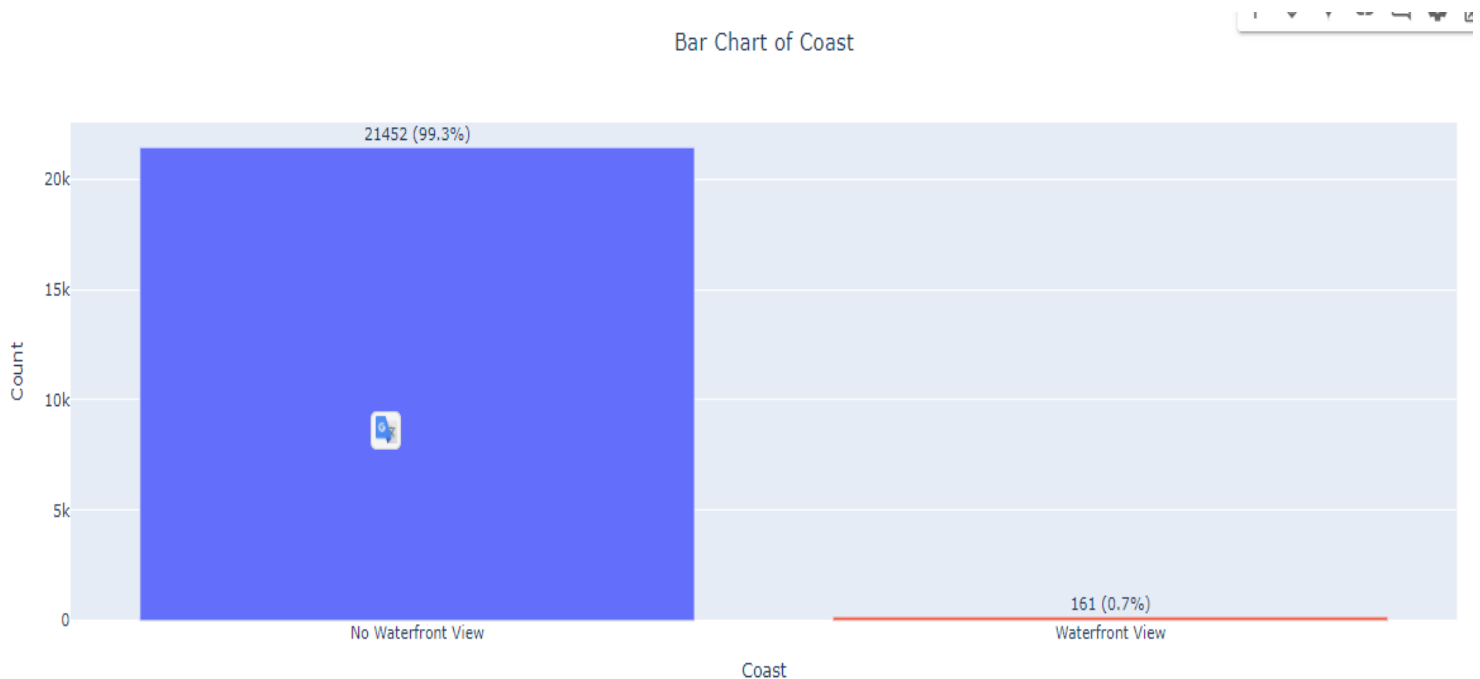
Renaming column Names - We have renamed some columns name using rename function.

```
Index(['cid', 'house_sold_date', 'price', 'bedrooms', 'bathrooms',  
      'living_measure', 'lot_measure', 'floors', 'coast', 'sight',  
      'condition', 'quality', 'ceil_measure', 'basement', 'yr_built',  
      'yr_renovated', 'zipcode', 'latitude', 'longitude', 'living_measure15',  
      'lot_measure15', 'furnished', 'total_area'],  
      dtype='object')
```

Exploratory Data Analysis –

Univariate analysis (Distribution and spread for every continuous attribute, distribution of data in categories for categorical ones).

Perform Univariate Analysis on Categorical Variable: -

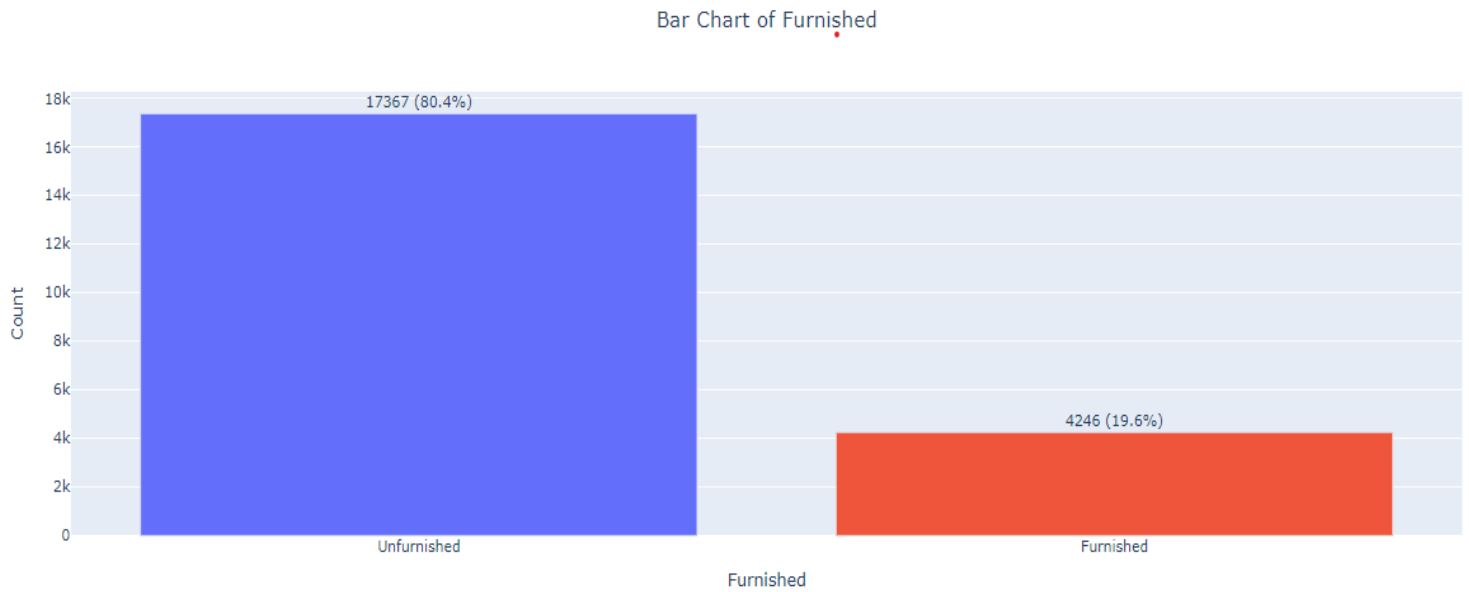


Coast Chart:

- No Waterfront View: 21,452 (99.3%) of the houses do not have a waterfront view.
- Waterfront View: 161 (0.7%) of the houses have a waterfront view.

Observation: Very few houses in the dataset have a waterfront view, making it a rare feature.

The data highlights that furnished houses and waterfront views are less common, potentially making them premium features. These attributes could influence pricing and buyer preferences significantly.

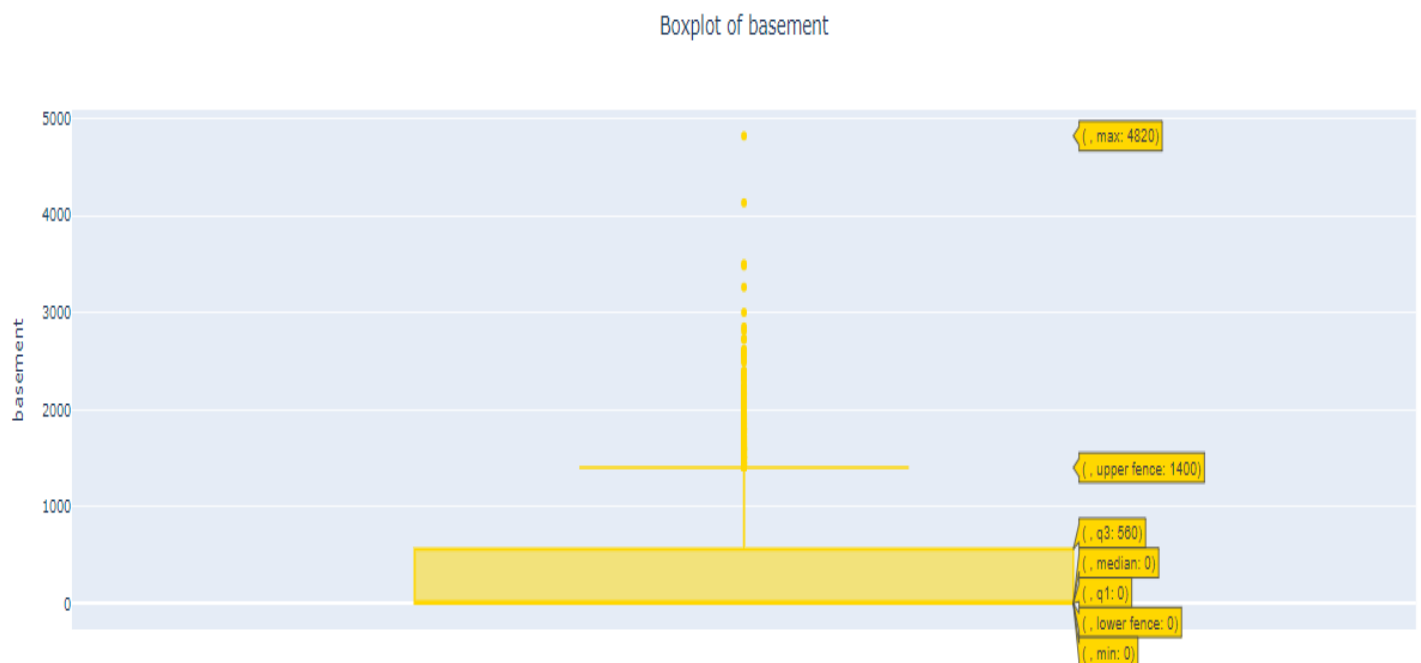


Furnished Chart:

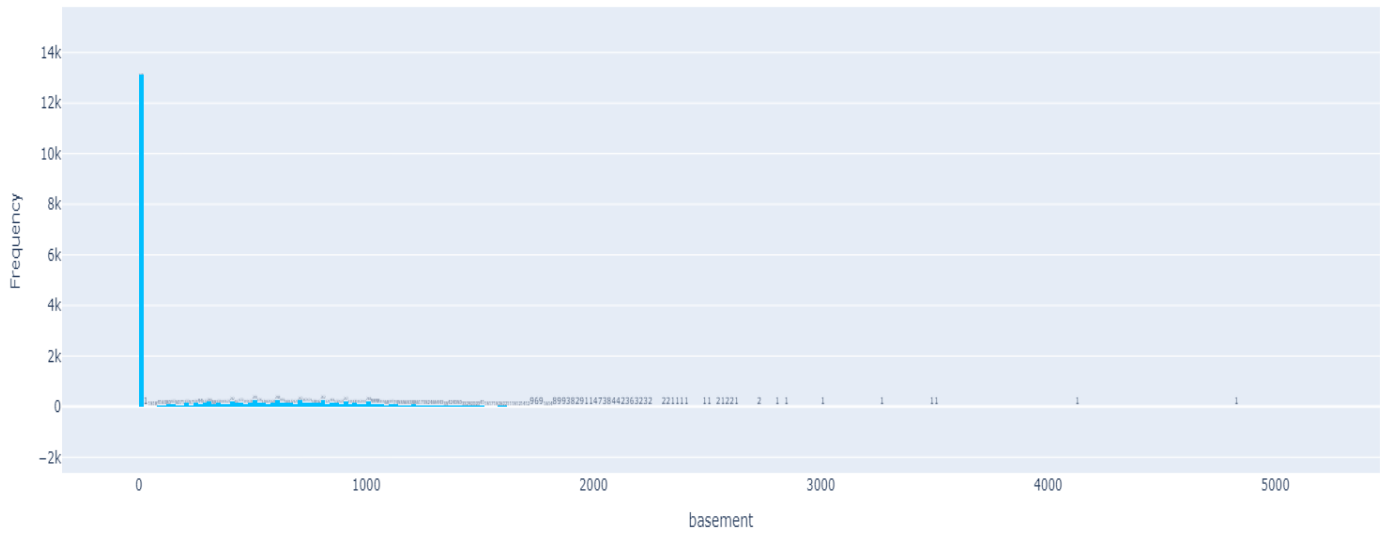
- Unfurnished: 17,367 (80.4%) of the houses are unfurnished.
- Furnished: 4,246 (19.6%) of the houses are furnished.

Observation: The majority of houses in the dataset are unfurnished, with only a small percentage being furnished.

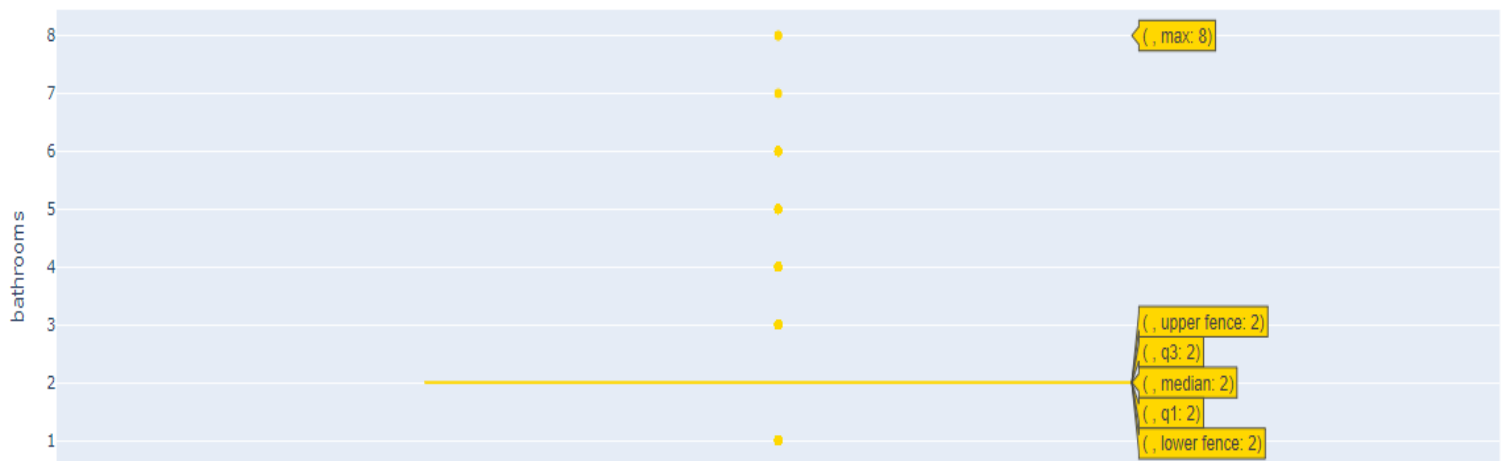
Perform Univariate Analysis on Continuous Variable -



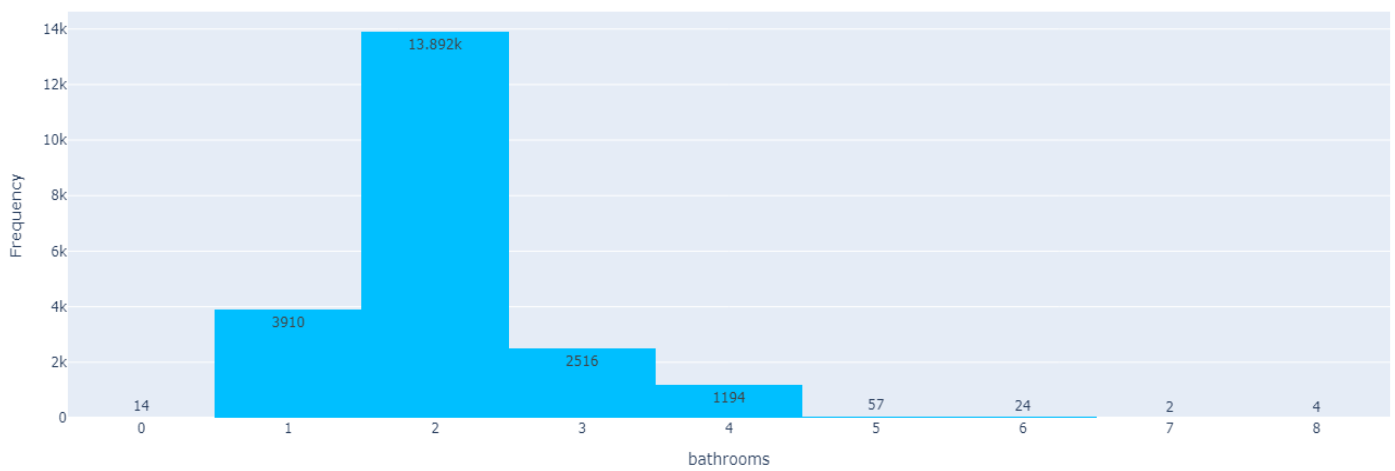
Histogram of basement



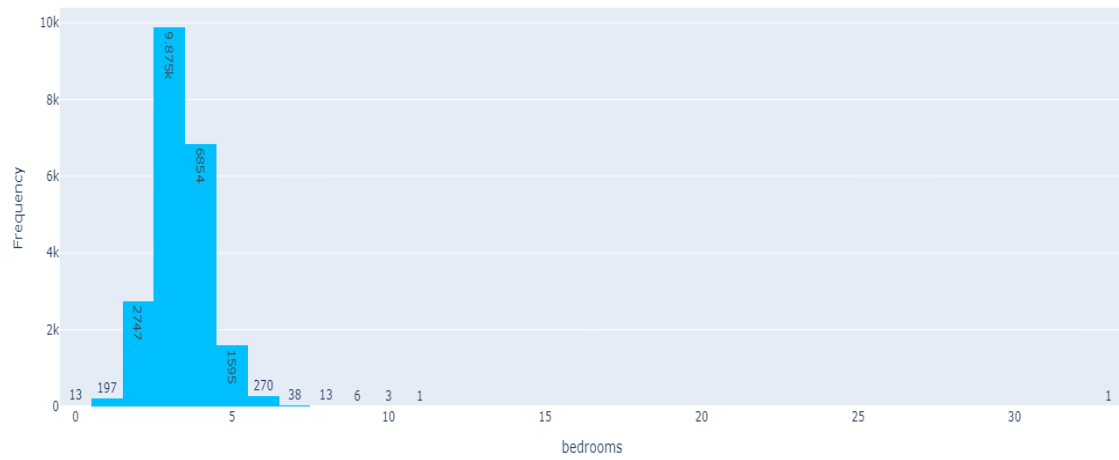
Boxplot of bathrooms



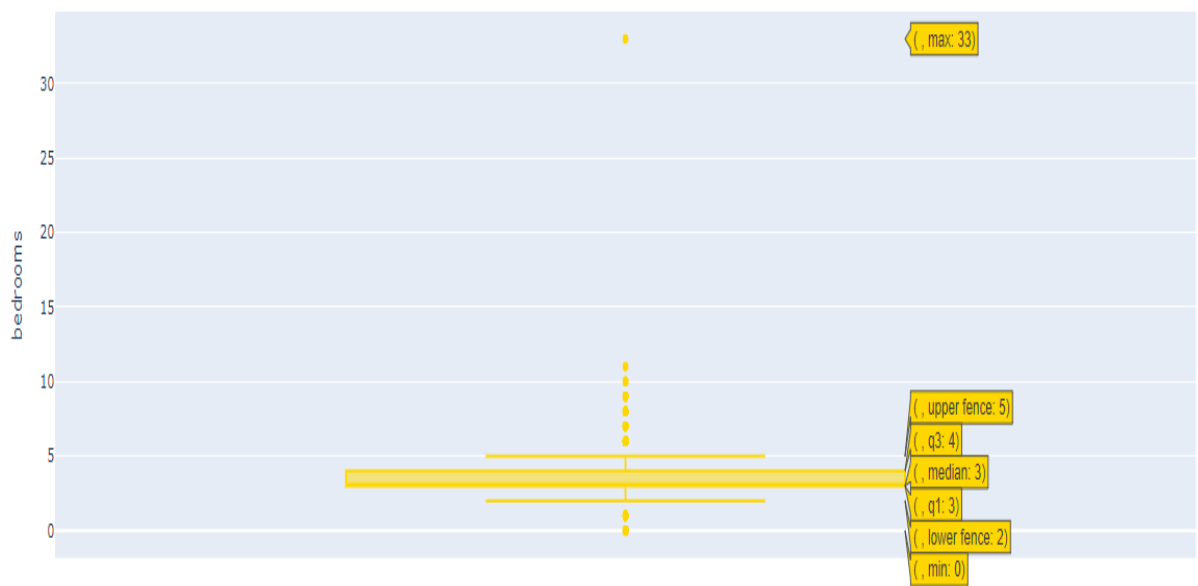
Histogram of bathrooms



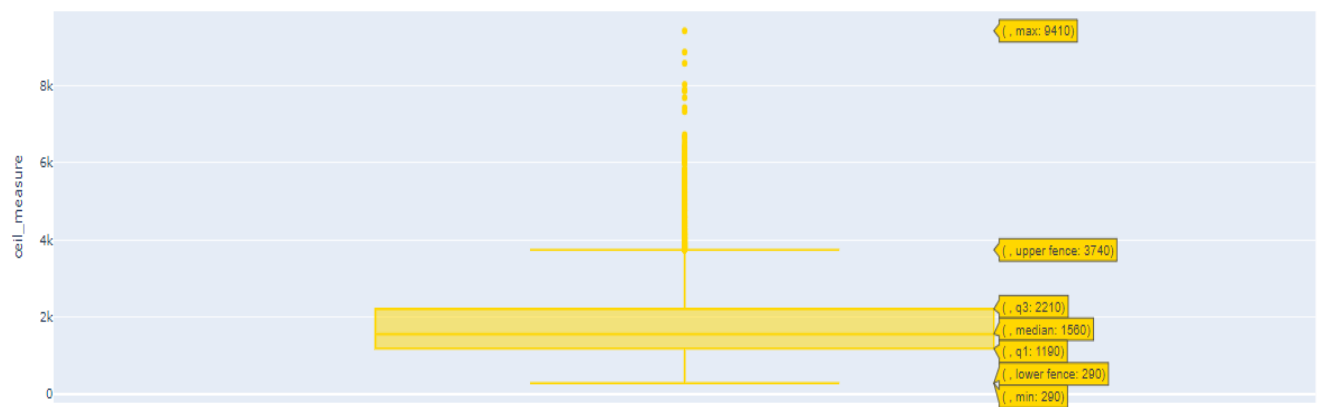
Histogram of bedrooms



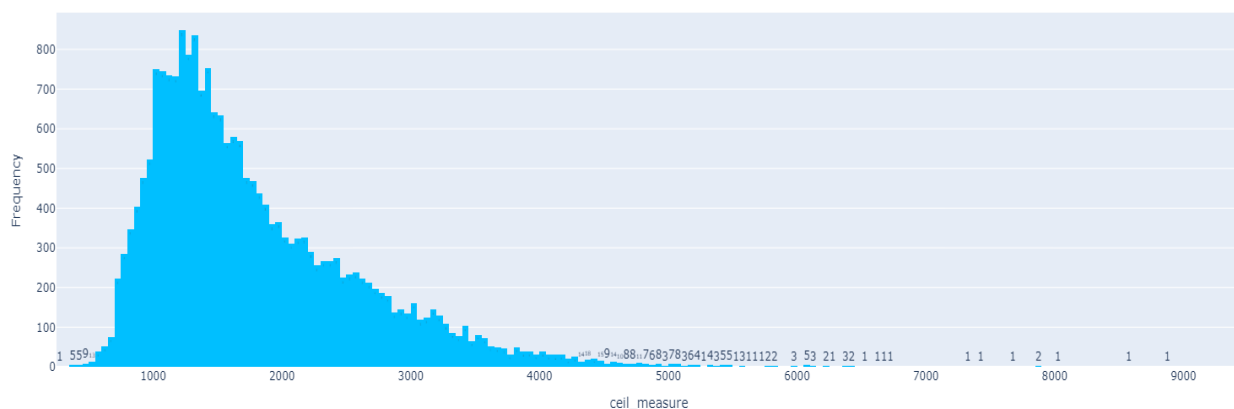
Boxplot of bedrooms



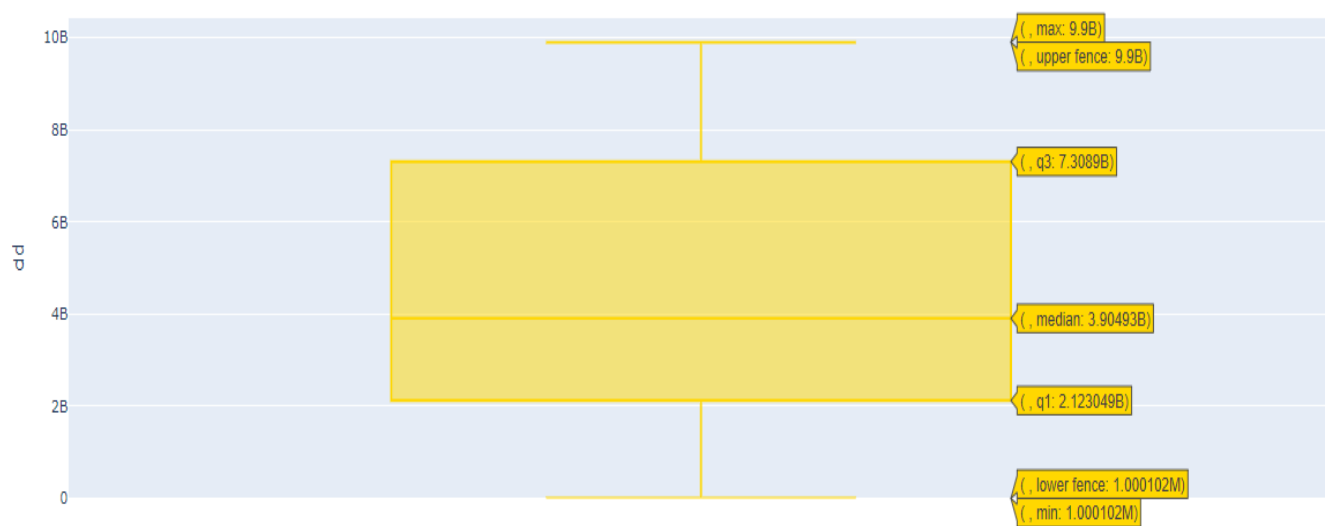
Boxplot of ceil_measure



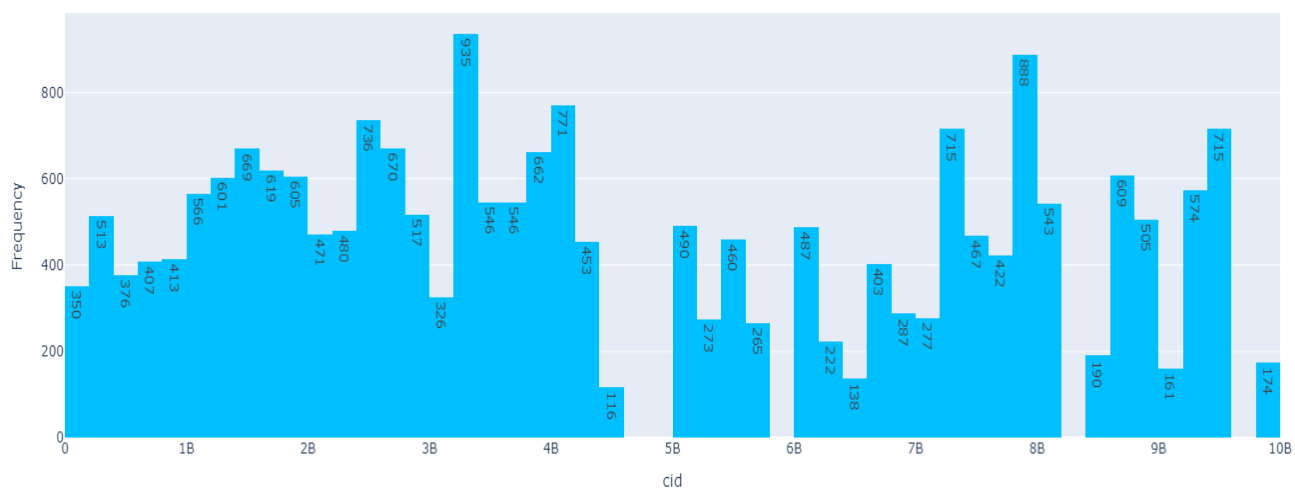
Histogram of ceil_measure

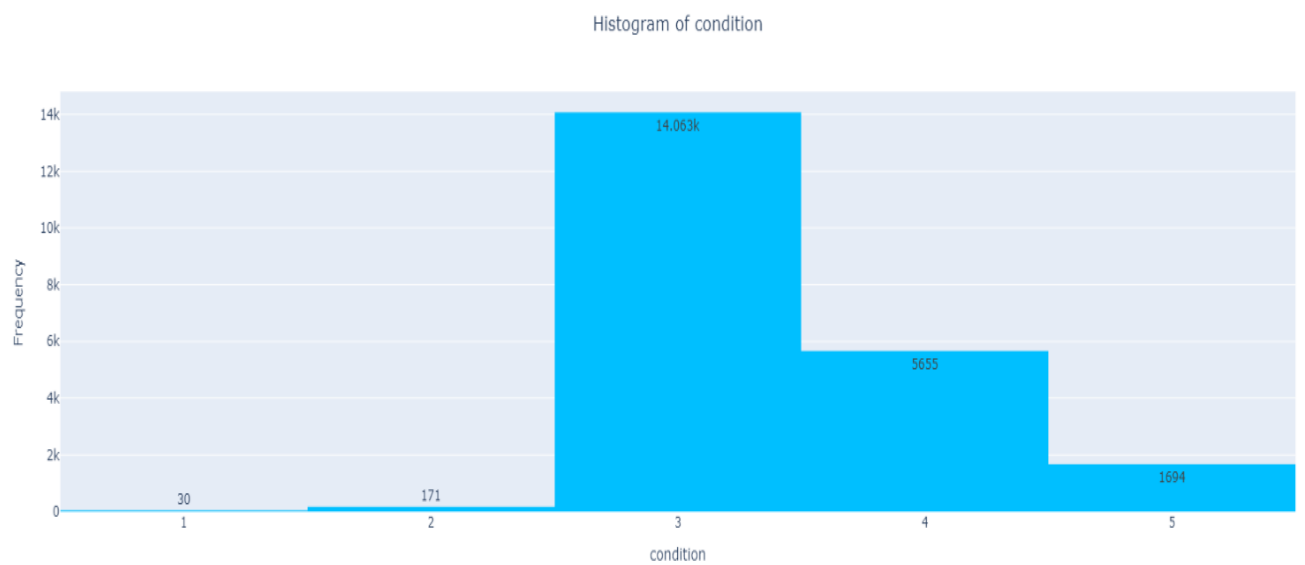
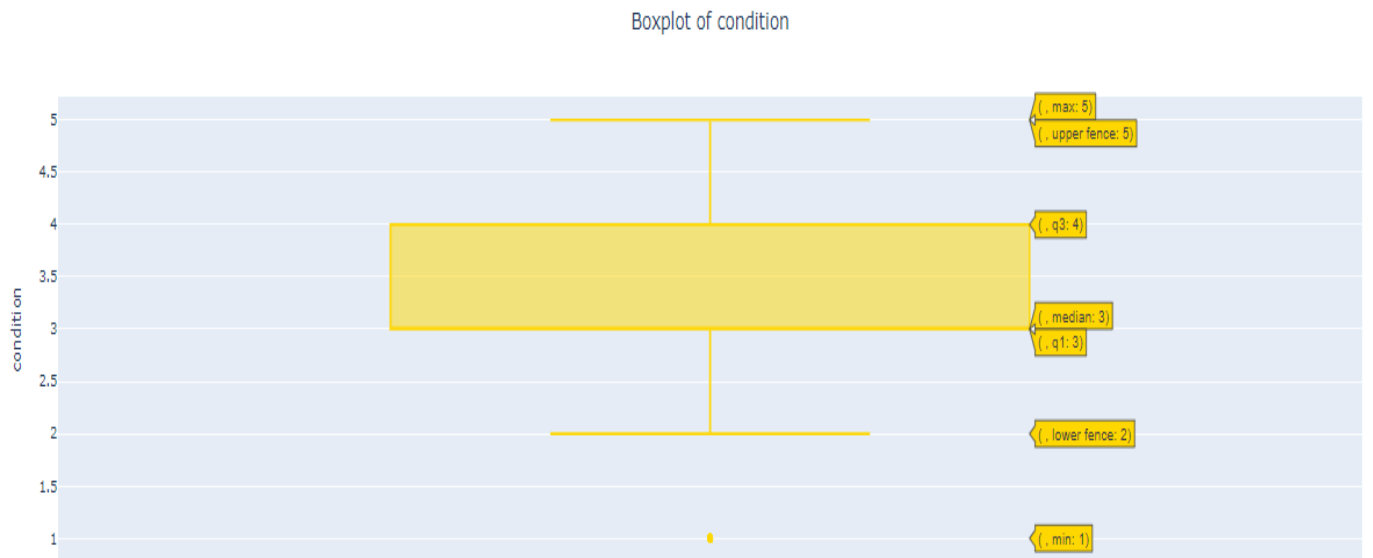


Boxplot of cid

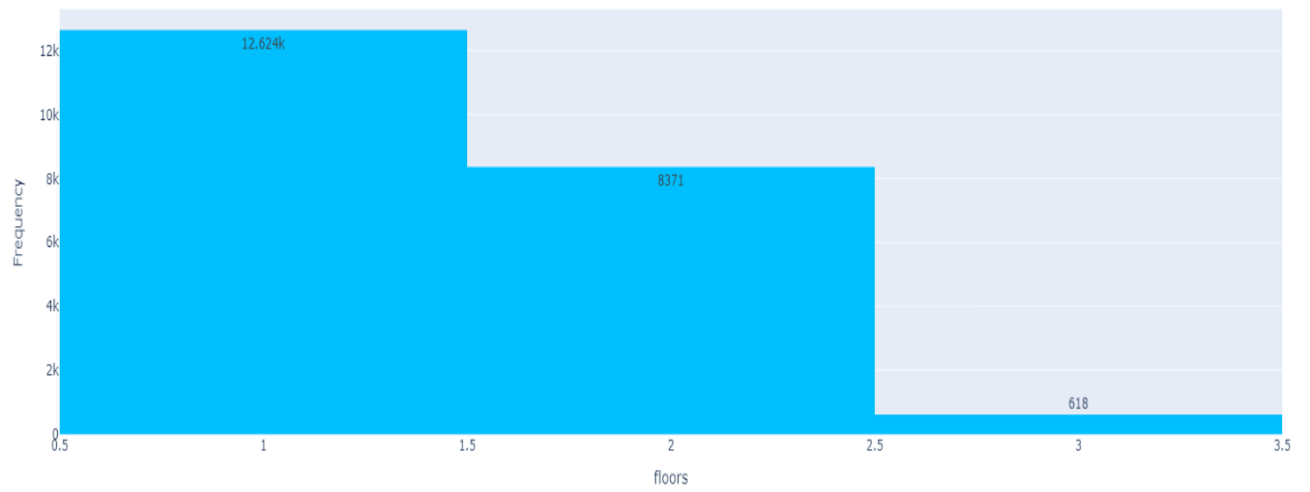


Histogram of cid

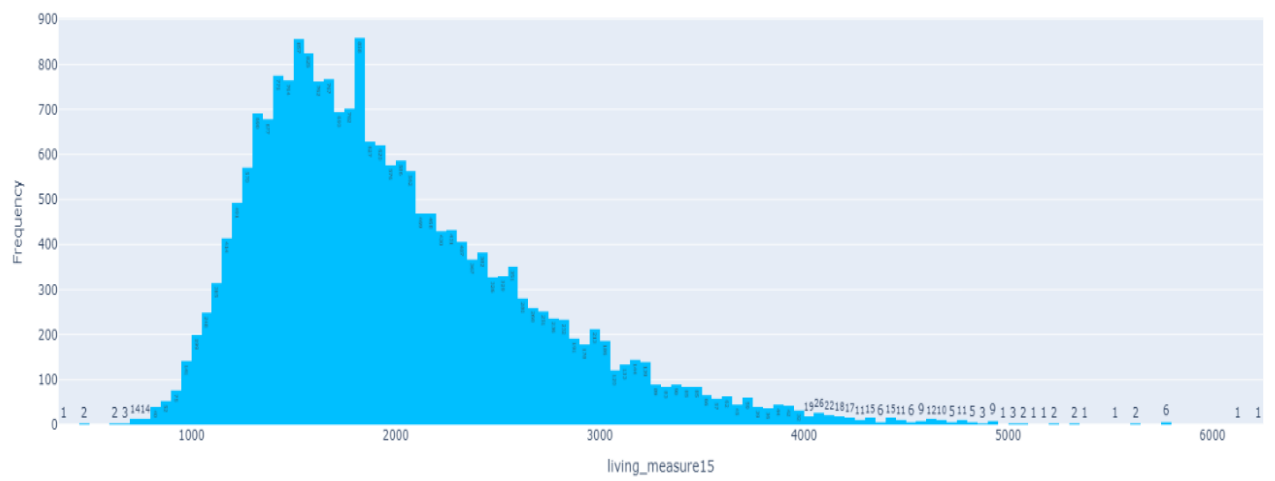




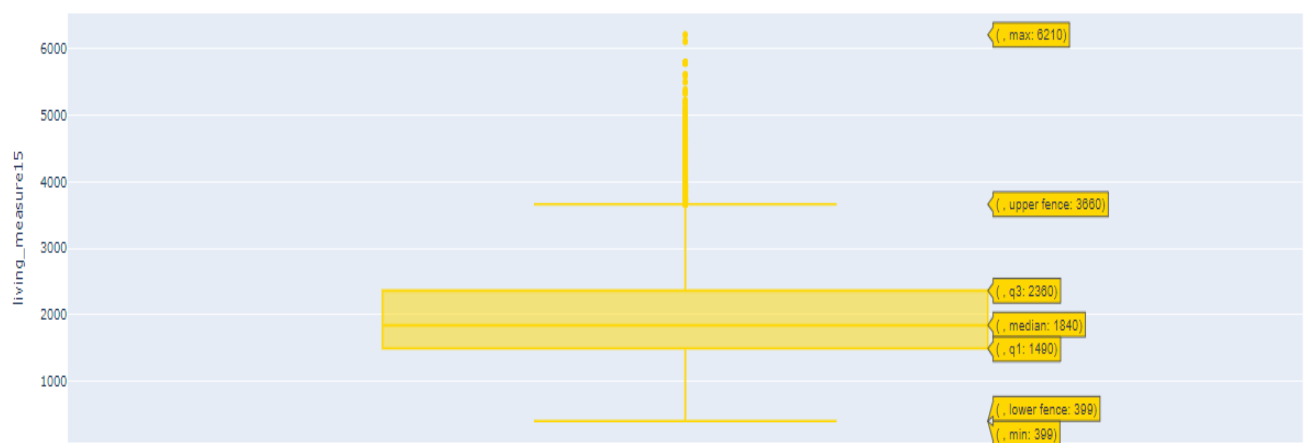
Histogram of floors



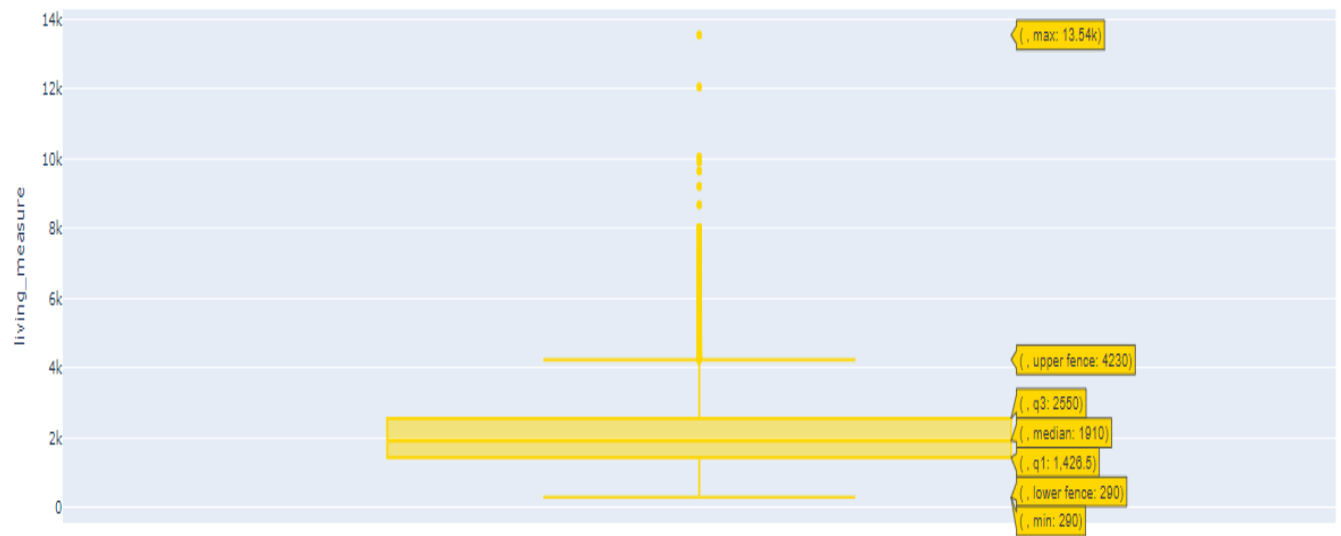
Histogram of living_measure15



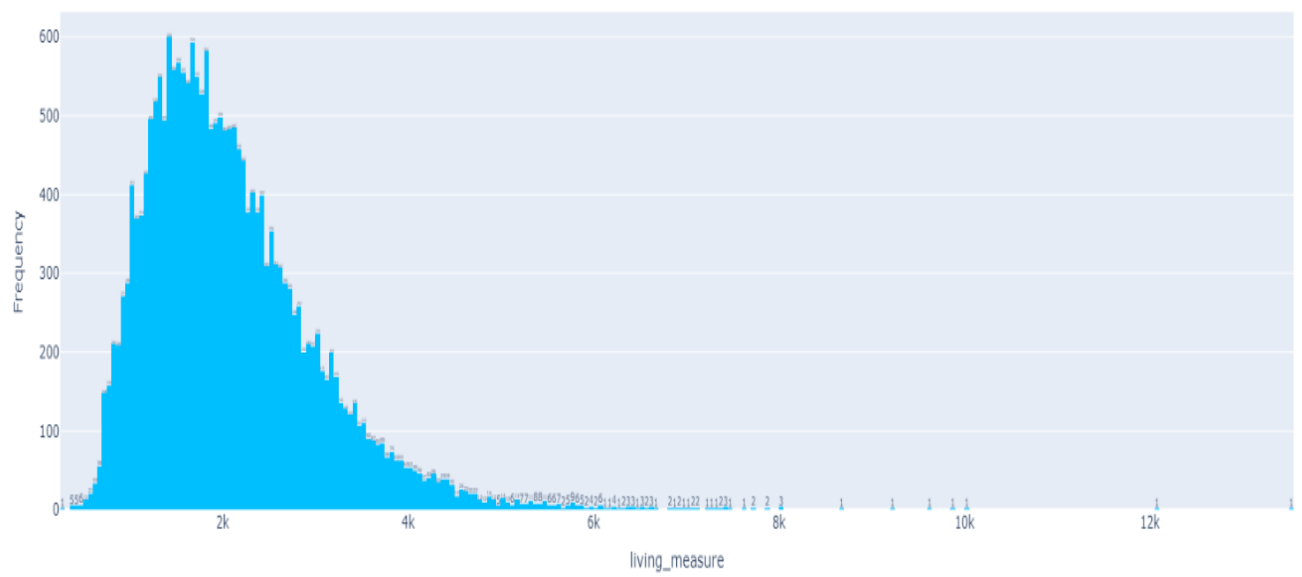
Boxplot of living_measure15



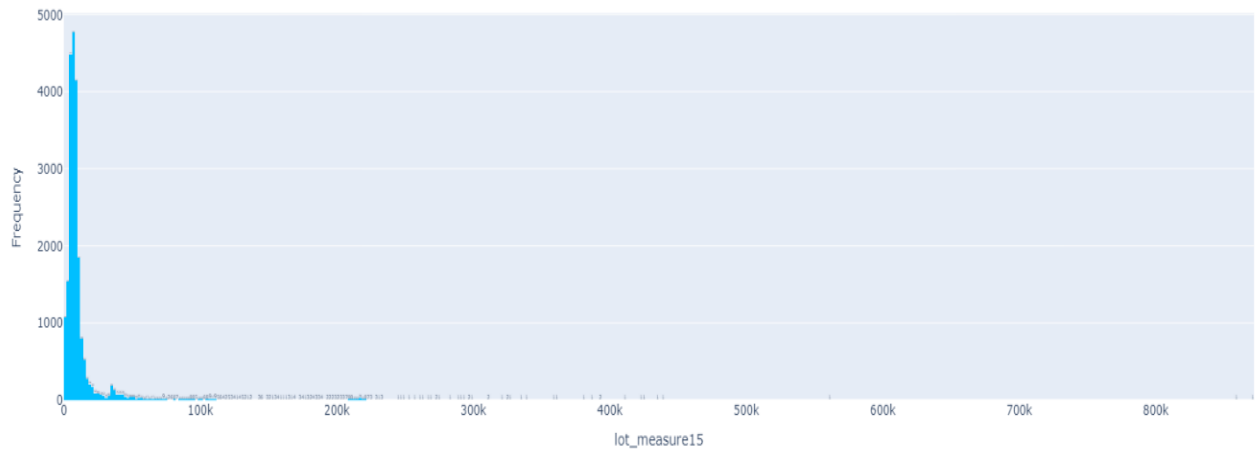
Boxplot of living_measure



Histogram of living_measure



Histogram of lot_measure15



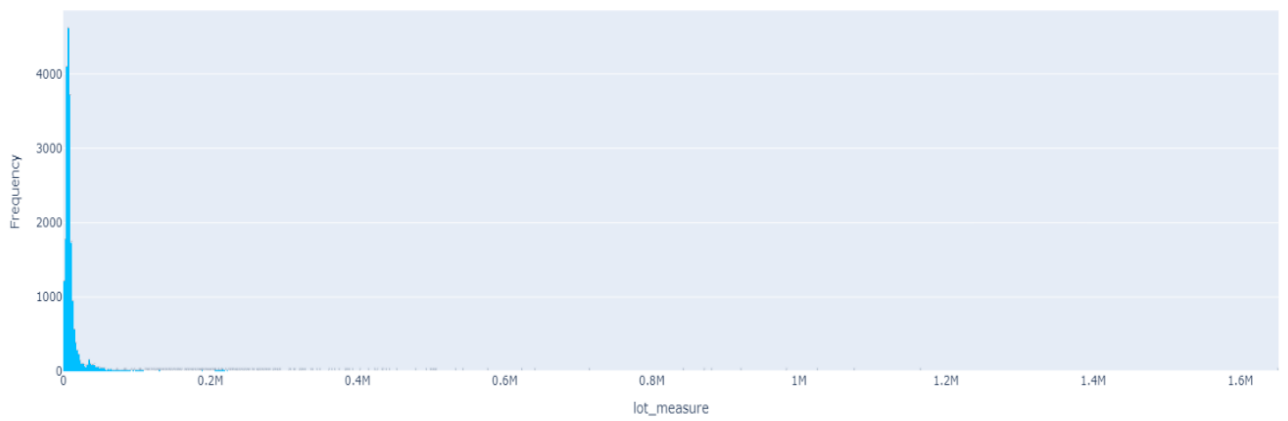
Boxplot of lot_measure15



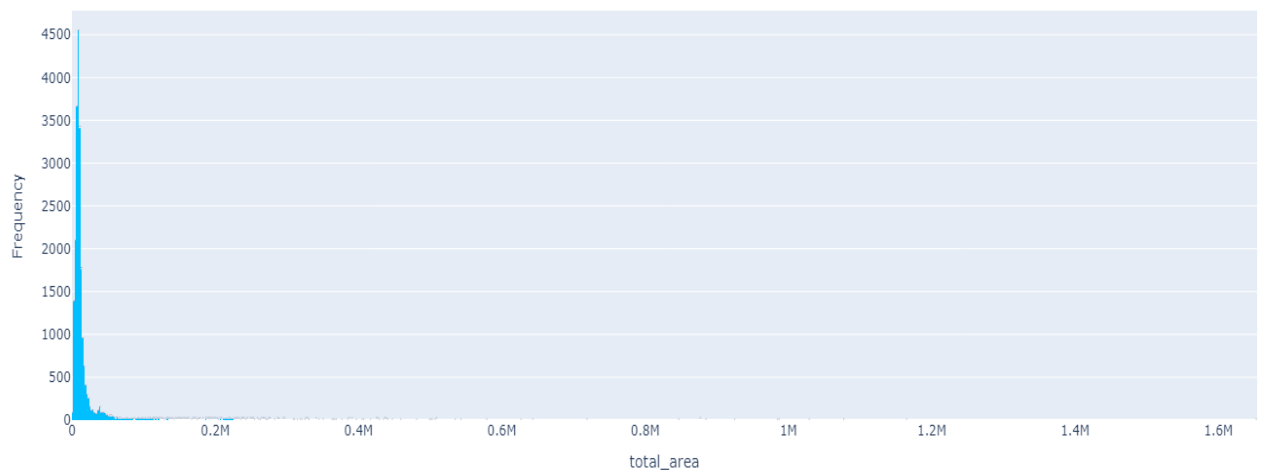
Boxplot of lot_measure



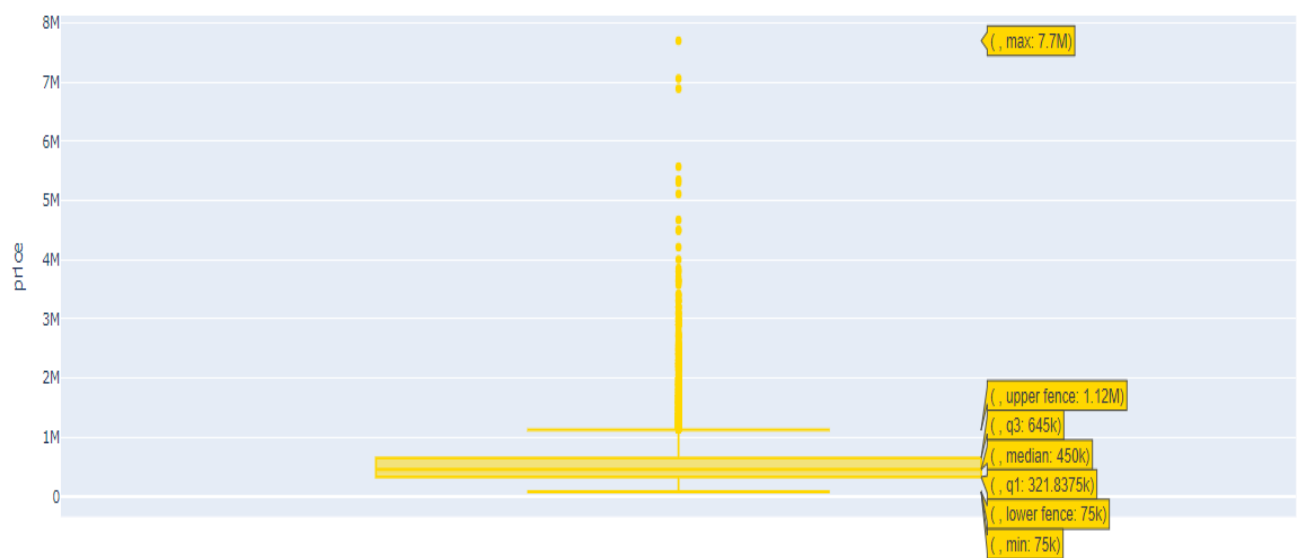
Histogram of lot_measure



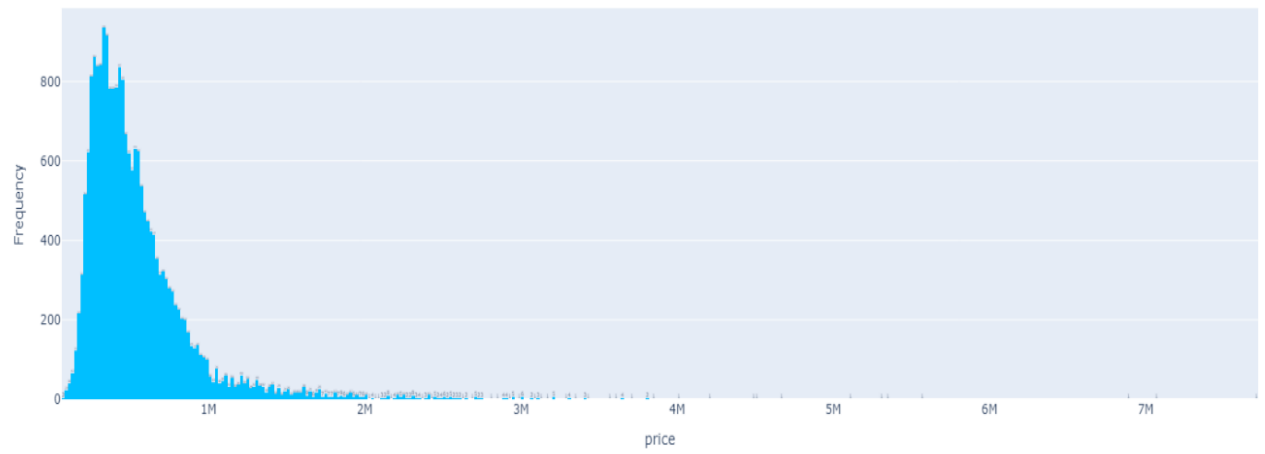
Histogram of total_area



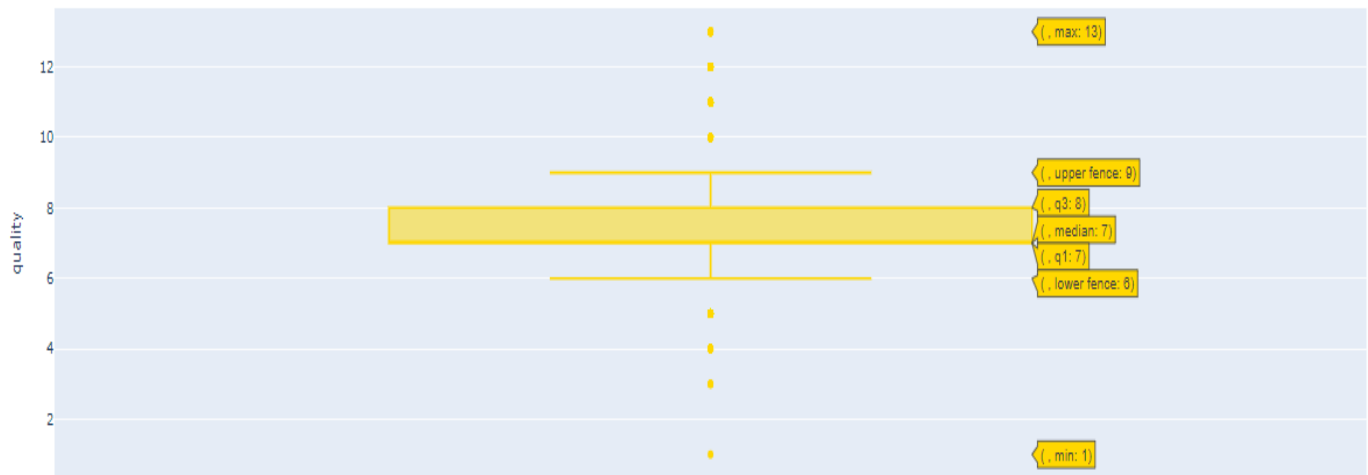
Boxplot of price



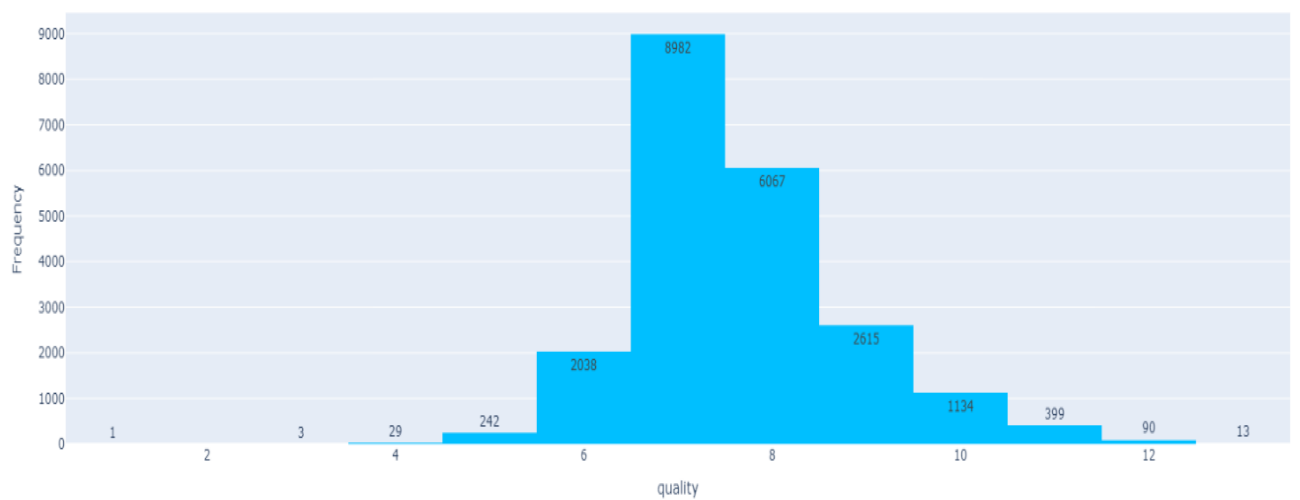
Histogram of price



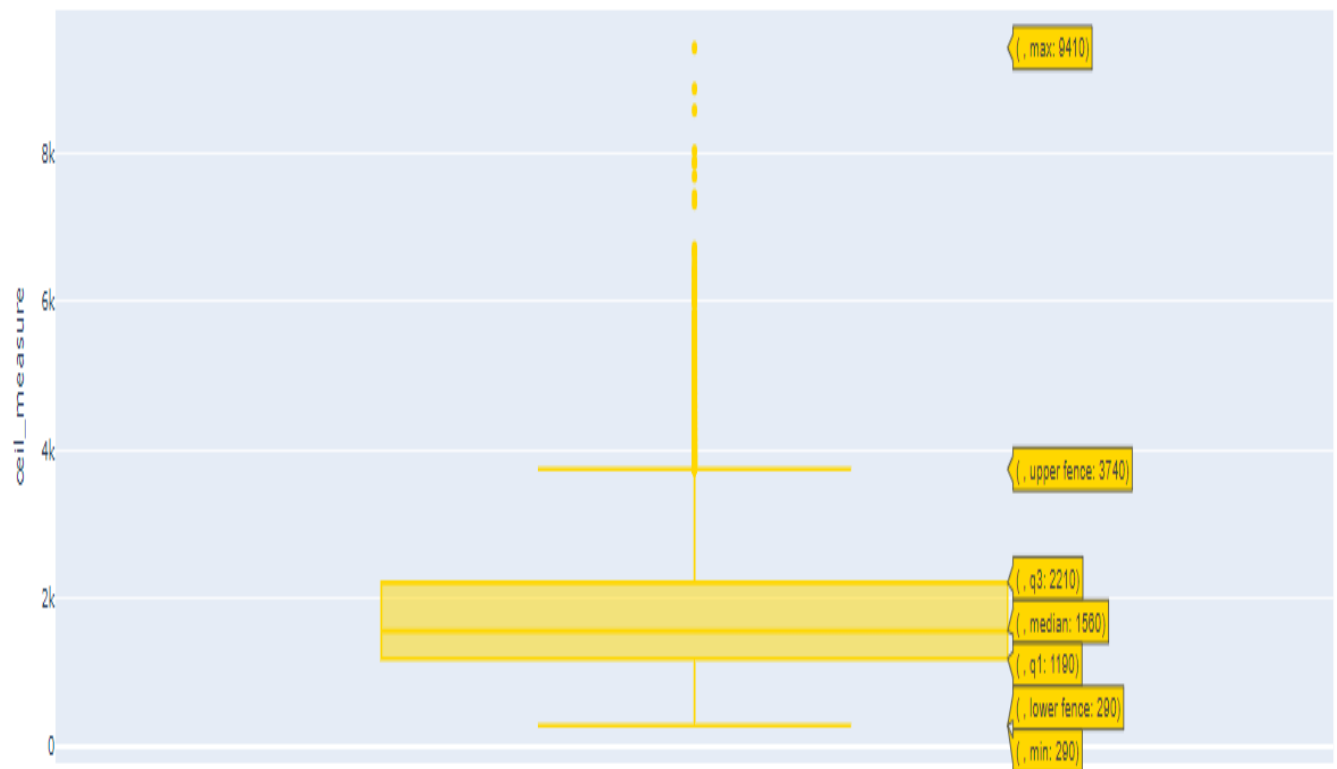
Boxplot of quality



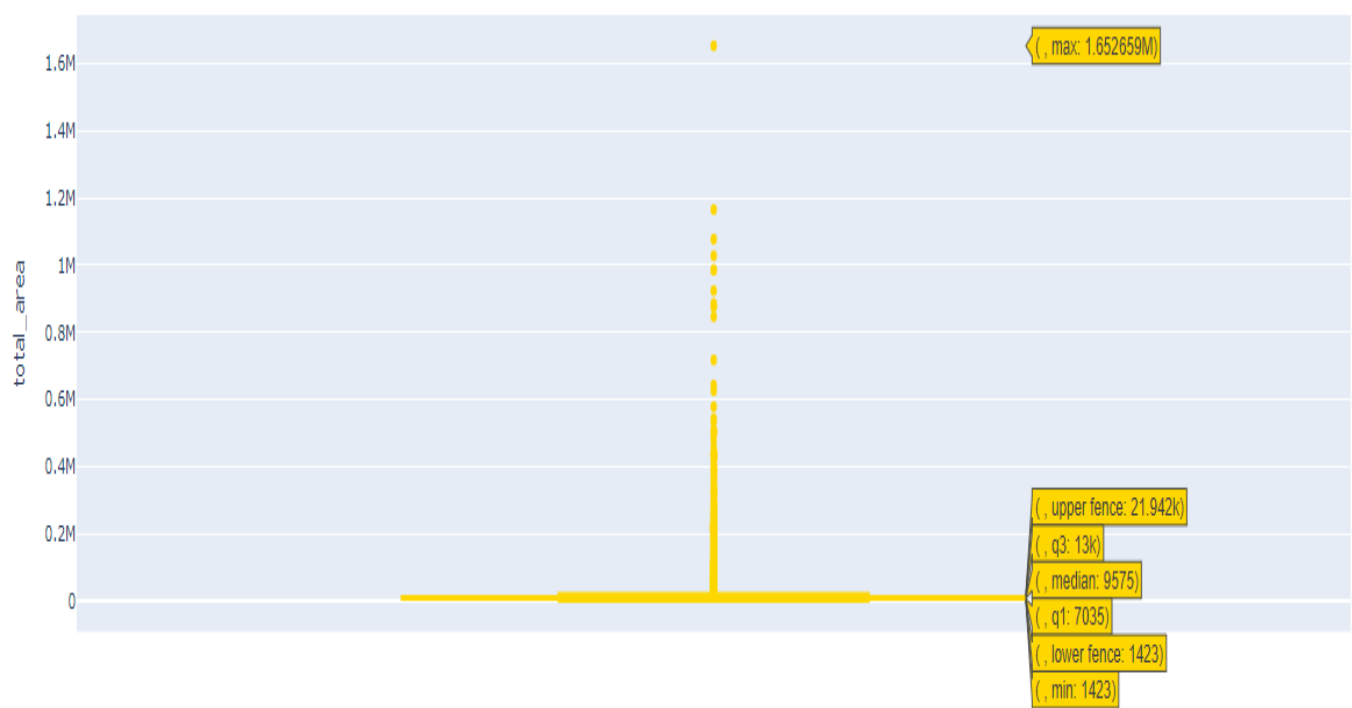
Histogram of quality



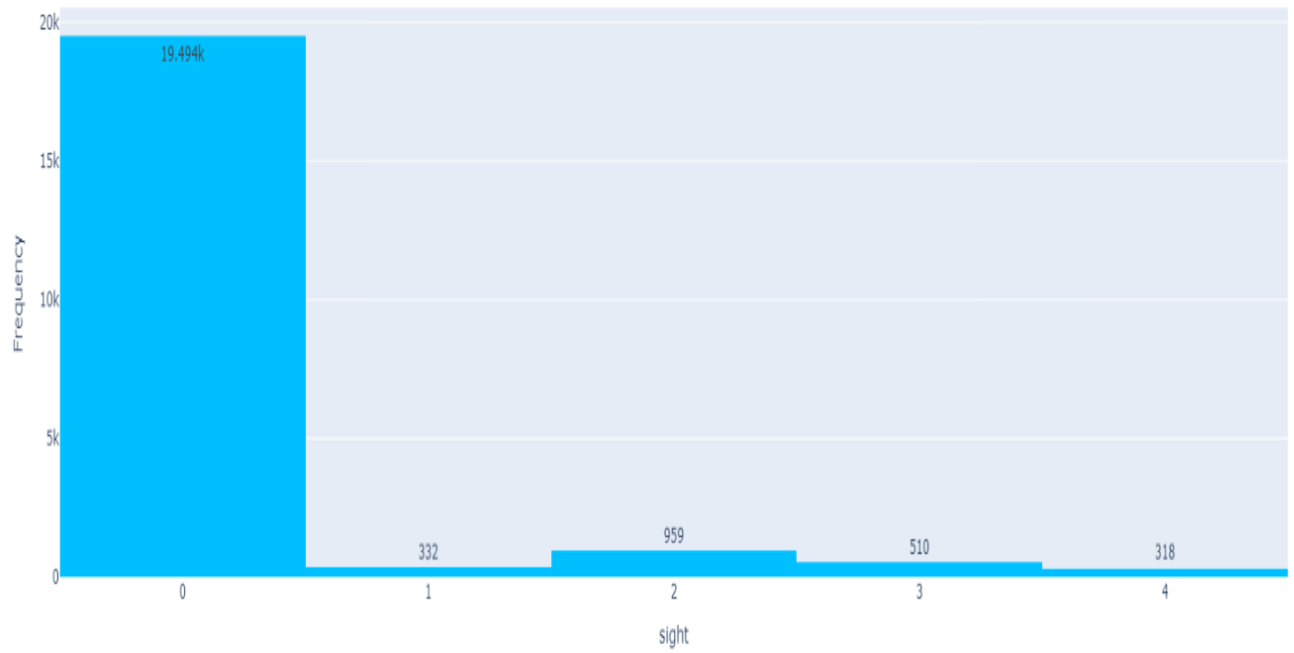
Boxplot of ceil_measure



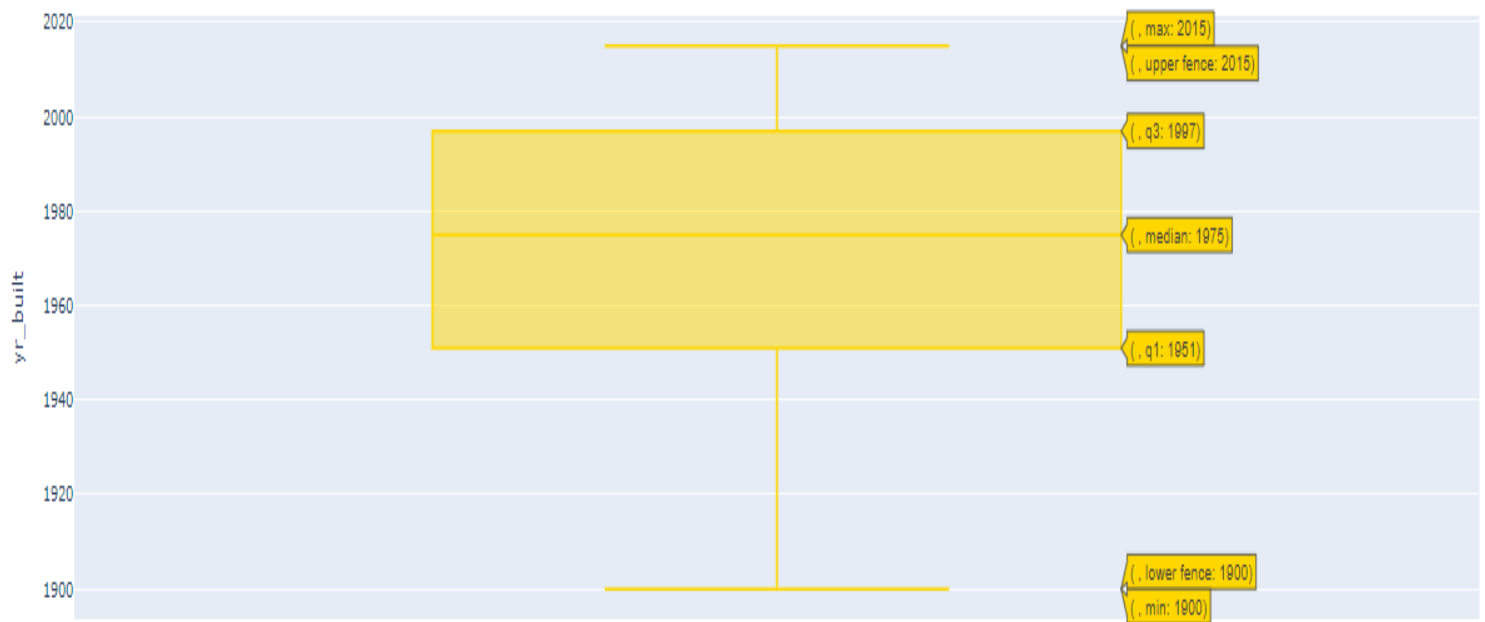
Boxplot of total_area



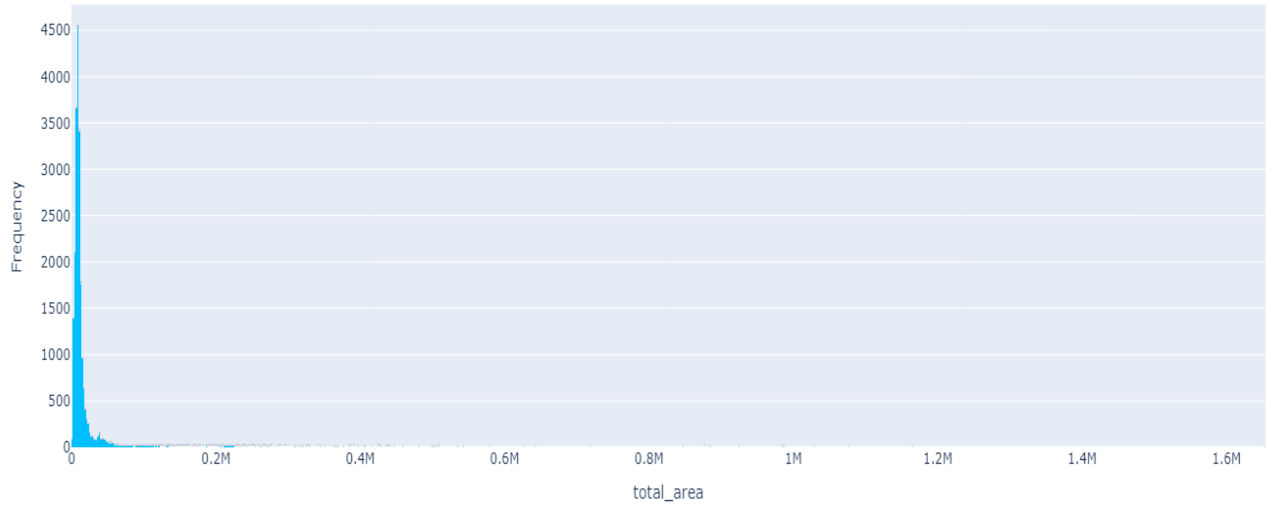
Histogram of sight



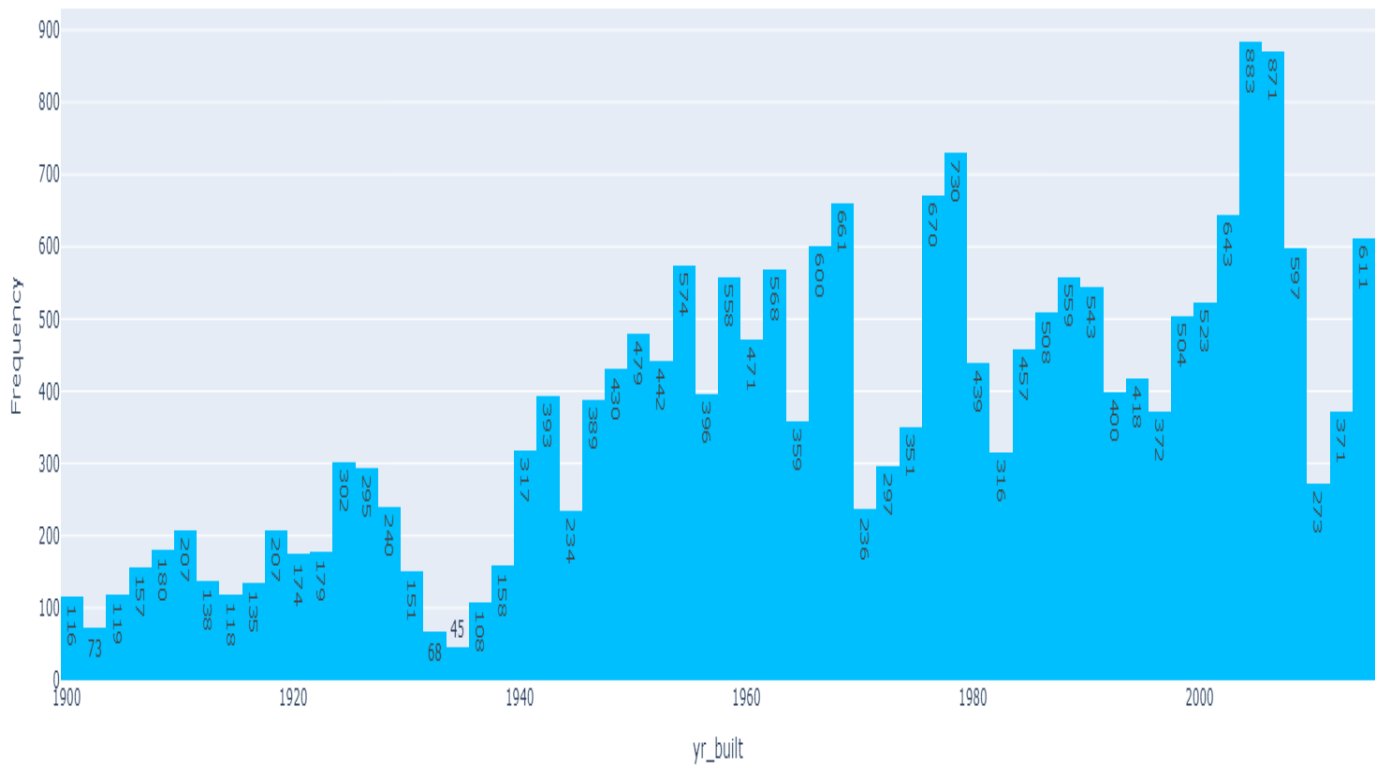
Boxplot of yr_built



Histogram of total_area



Histogram of yr_built



Observations from the visualizations provided:

1 - Basement

Boxplot of Basement: -

- * The data is highly skewed, with the majority of values clustered near zero.
- * The lower fence, minimum, median, and first quartile are all at 0, indicating a significant proportion of the data has no basement area (0 value).
- * The upper fence is at 1400, but there are several outliers extending up to 4820.
- * The interquartile range (IQR) is minimal, showing limited variability within the central portion of the data.

Histogram of Basement: -

- * A high frequency of zeros dominates the data.
- * Beyond the 0 value, frequencies drop sharply, with a long tail indicating a few properties with large basement areas.
- * This distribution is strongly right-skewed, with outliers influencing the range.

2 - Bathrooms

Boxplot of Bathrooms:

- * The data has a maximum value of 8, but the upper fence is at 2, indicating outliers for properties with more than 2 bathrooms.
- * The median is at 2, showing that most properties have 2 bathrooms.

Histogram of Bathrooms: -

- * The majority of properties (around 13.9k) have 2 bathrooms, followed by properties with 1 bathroom (~3.9k).
- * A smaller number of properties have more than 3 bathrooms, with very few going up to 6, 7, or 8 bathrooms.

3 - Bedrooms: -

Histogram of Bedrooms:

- * Most properties have 3 or 4 bedrooms, with a peak around 9.87k at 3 bedrooms and 6.85k at 4 bedrooms.
- * Properties with fewer (1–2) or more (5+) bedrooms are less common.
- * A few extreme outliers exist (e.g., properties with 15, 20, or even 30 bedrooms).

Boxplot of Bedrooms:

- * 1 - Median: The median value is at 3 bedrooms, indicating that 50% of the properties have 3 bedrooms or fewer, and the other 50% have more than 3 bedrooms.
- * 2 - Interquartile Range (IQR): The first quartile (Q1) is at 3, and the third quartile (Q3) is at 4, indicating that the middle 50% of the data ranges between 3 and 4 bedrooms. This shows that most properties have between 3 to 4 bedrooms.
- * 3 - Whiskers:
 - * The lower whisker extends to 2 bedrooms, suggesting that the smallest non-outlier value is 2 bedrooms.
 - * The upper whisker extends to 5 bedrooms, indicating that most properties have no more than 5 bedrooms, except for outliers.
- * 4 - Outliers:
 - * There is a significant outlier at 33 bedrooms, which is unusually high compared to the rest of the dataset.
 - * Such an outlier might represent a unique or special property and could impact the analysis if not handled appropriately.
- * 5 - Minimum and Maximum: The minimum value in the dataset is 0 bedrooms, which could represent properties like studio apartments or listings without specified bedroom data. The maximum, excluding outliers, is 5 bedrooms.

4 - Ceil Measure:

Boxplot of ceil_measure: -

- * The median of ceil_measure is around 1560.
- * The interquartile range (IQR) is from approximately 1180 to 2210.
- * There are outliers extending up to the maximum value of 9410.
- * The upper fence is at 3740, which indicates that any value above this is considered an outlier.
- * The lower whisker is at 290, with the minimum value being 290.

Histogram of ceil_measure:

- * The data is right-skewed, with most values concentrated between 500 and 3000.
- * There is a long tail extending towards higher values, indicating a few very large ceil_measure values (outliers).

* The highest frequency of data points falls between 1000 and 1500, suggesting that this range is the most common for ceil_measure.

5 - Condition: -

Boxplot of Condition:

Median: - The median value is 3, which means that half of the data points have a condition score less than or equal to 3.

Interquartile Range (IQR): - The IQR spans from 3 (Q1) to 4 (Q3), indicating that the middle 50% of the data lies between these values.

Range: - The data ranges from a minimum of 1 to a maximum of 5.

Outliers: - There is one outlier at the lower end with a condition score of 1.

Histogram of Condition:

The histogram shows that the majority of the data points fall in the condition score of 3, with the highest frequency at around 14,063.

The condition score of 4 has the second-highest frequency, with 5,655 occurrences.

Condition scores of 1 and 2 have very low frequencies, with 30 and 171 occurrences, respectively.

Condition score 5 also has a significant number of occurrences, with 1,694 data points.

6 - Floors: -

Boxplot of Floors: -

* The minimum number of floors is 1.

* The maximum number of floors is 3.

* The median (Q2) is approximately 1.5, indicating that 50% of the data points have 1.5 floors or fewer.

* The interquartile range (IQR) is between 1 and 2 floors.

* There are no apparent outliers, as the lower and upper fences match the minimum and maximum values.

Histogram of Floors: -

* The most common number of floors is 1, with a frequency of approximately 812.624k.

* The second most common number of floors is 2, with a frequency of 8.371k.

* Very few houses have 3 floors, with a frequency of 618.

* The distribution is skewed towards fewer floors, with a significant drop in frequency as the number of floors increases.

7 - Living Measure (living_measure15): -

Histogram of living_measure15: - The distribution of living_measure15 is right-skewed, with most values concentrated between 1000 and 2500 square feet. There is a gradual decrease in frequency as the size increases, with a few outliers beyond 4000 square feet.

Boxplot of living_measure15: - The median living space is approximately 1840 square feet. The interquartile range (IQR) spans from around 1490 to 2380 square feet. There are several outliers beyond the upper fence (approximately 3860 square feet), indicating a few homes with significantly larger living spaces.

8 - Lot Measure (lot_measure15) -

Histogram of lot_measure15: - The distribution of lot_measure15 is highly right-skewed, with most values concentrated below 20,000 square feet. A few extreme values extend beyond 200,000 square feet, indicating large lot sizes in some cases.

Boxplot of lot_measure15: - The median lot size is around 7820 square feet, with the IQR ranging from approximately 5100 to 10,080 square feet. There are several outliers beyond the upper fence (approximately 17,550 square feet), suggesting a small number of properties with significantly larger lot sizes.

9 - Living Measure: -

Boxplot of living_measure: -

* The minimum value is around 290, and the maximum value reaches approximately 13,540, with a large range.

* The median value is about 1,910, indicating that half of the data points lie below this value.

* The lower quartile (Q1) is around 1,426.5, and the upper quartile (Q3) is approximately 2,590.

* There are outliers above the upper fence, indicating some very large values in the dataset.

* The spread of the data is quite large, with a noticeable presence of outliers that extend significantly beyond the upper fence.

Histogram of living_measure: -

* The distribution is right-skewed, with most values clustered between 1,000 and 3,000.

* The frequency decreases as the living measure increases, with very few properties having a measure greater than 8,000.

* The histogram shows a long tail towards the right, indicating the presence of a few very large values (outliers).

11- Lot Measure: -

Boxplot of lot_measure: -

* The minimum value is around 520, and the maximum value extends to approximately 1,651,359, showing an extensive range.

* The median value is about 7,818, which is significantly lower than the maximum value, indicating a skewed distribution.

* The lower quartile (Q1) is around 5,040, and the upper quartile (Q3) is approximately 10,828.5.

* There are a significant number of outliers above the upper fence, which indicates extreme values in the dataset.

* The data shows a vast spread, and the presence of outliers is quite pronounced, with some values being extraordinarily high.

Histogram of lot_measure: - * The distribution is highly right-skewed, with most values concentrated at the lower end of the scale.

* The frequency drops rapidly as the lot measure increases, with very few properties having a measure greater than 100,000.

* The histogram shows a sharp decline after the initial peak, with a long tail towards the right, indicating a few extreme values (outliers).

12 - Price:

Boxplot of Price: -

* Median Price: The median price is around \$450,000, which provides a central tendency of the data.

* Interquartile Range (IQR): The IQR (between Q1 and Q3) spans from approximately \$321,837 to \$645,000, indicating a moderate spread of prices within the middle 50% of the data.

* Upper Fence: The upper fence is at \$1.12M, above which data points are considered potential outliers.

* Outliers: There are numerous outliers beyond the upper fence, with the maximum price reaching \$7.7M, suggesting the presence of very high-end properties.

* Minimum Price: The minimum price observed is \$75,000, indicating the lower end of the market.

Histogram of Price: -

* Distribution Shape: The histogram shows a right-skewed distribution, with most properties priced below \$1M.

* Frequency Peak: The highest frequency of prices is around the lower end, with the mode being below \$1M.

* Long Tail: There is a long tail extending to higher prices, which aligns with the presence of outliers in the boxplot.

* Rare High Prices: Prices above \$3M are rare, which is consistent with the outliers seen in the boxplot.

13 - Quality:

Boxplot of quality: -

* The median quality value is 7.

* The interquartile range (IQR) is between 6 (Q1) and 8 (Q3).

* The data has outliers on both ends, with the minimum outlier at 1 and the maximum at 13.

* The overall spread of the data is between 1 and 13.

Histogram of quality: -

* The distribution of quality is skewed to the right.

* Most of the data points fall between 6 and 8.

* The highest frequency is at quality value 6, with a frequency of around 9,000.

* There are very few data points at the lower end (1-4) and the higher end (10-13).

14 - Sight:

Boxplot of sight: -

* The sight values mostly range between 0 and 1, with the majority being at 0.

* The interquartile range (IQR) lies between 0 and 1.

* There are several outliers present, with values going up to 4.

Histogram of sight: -

* The distribution is heavily skewed towards 0, indicating that most data points have a sight value of 0.

* There is a sharp decline in frequency as the sight value increases.

* Few data points have higher sight values, with a few outliers reaching up to 4.

15 - Total Area: -

Boxplot of total_area: -

Minimum Value: The minimum total area is 1423 units.

Lower Fence: The lower fence is also at 1423 units, indicating there are no lower outliers.

First Quartile (Q1): 7035 units, indicating that 25% of the data is below this value.

Median: 9575 units, meaning half of the data is below this value.

Third Quartile (Q3): 13,000 units, indicating that 75% of the data is below this value.

Upper Fence: 21,942 units, suggesting any value above this might be considered an outlier.

Maximum Value: The maximum total area is approximately 1.65 million units, which is a significant outlier compared to the majority of the data.

Histogram of total_area:

* The data distribution is highly skewed to the right, with most of the total area values concentrated near the lower end (between 0 and 200,000 units).

* The frequency of total area values drastically decreases as the area size increases, with very few instances of total area above 1 million units.

16 - Year Built

Boxplot of yr_built: -

* The minimum year built is 1900, and the maximum is 2015.

* The median year built is 1975, with the first quartile (Q1) around 1961 and the third quartile (Q3) around 1997.

* There are no apparent outliers as the data ranges from 1900 to 2015, and all points fall within the lower and upper fences.

Histogram of yr_built: -

- * The distribution is somewhat right-skewed, indicating that a larger number of houses were built in more recent years.
 - * The frequency increases notably in the years around 2000, showing a peak in that period.
 - * There are fewer houses built before 1940 and a gradual increase in the number of houses built from 1950 onwards, with peaks around the 1970s and the early 2000s.
-

Key Observations -

1 - Basement Data: -

- * Many properties lack a basement.
- * There are outliers that indicate a few properties have very large basement areas, significantly distorting the scale.

2 - Bathroom Data:

- * Most properties have 2 bathrooms, with very few deviations.
- * A few properties with an unusually high number of bathrooms (up to 8) form outliers.

3 - Bedrooms:

- * The data is concentrated around 3 to 4 bedrooms, which seems typical for most properties in this dataset.
- * The presence of an extreme outlier (33 bedrooms) suggests there is a unique property or possible data entry error that may need further investigation.
- * Handling this outlier appropriately is crucial for accurate modeling and analysis, especially in predictive tasks.

4 - Ceil Measure:

The key observation is that the ceil_measure data is right-skewed with a concentration of values between 1000 and 1500, while there are significant outliers on the higher end, extending up to 9410. This suggests that most properties have moderate ceil_measure values, but a few have exceptionally high values.

5 - Condition:

- * The majority of the data points are concentrated around condition scores of 3 and 4, indicating that most items are in fair to good condition.
- * Very few items have a condition score of 1 or 2, suggesting that poor condition items are rare in the dataset.

6 - Floors:

- * Most of the properties have 1 or 2 floors, indicating that single and two-story homes are the most common.
- * Properties with 3 floors are quite rare.
- * There is a clear preference for properties with fewer floors, possibly due to regional architectural trends or zoning laws.

7 - living_measure15:

1 - **Distribution:** The distribution of living_measure15 is right-skewed, indicating that most properties have a living area size between 1000 and 2500 square feet.

2 - **Central Tendency:** The median living area size is approximately 1840 square feet, suggesting that half of the properties have living spaces below this value.

3 - **Spread:** The interquartile range (IQR) spans from about 1490 to 2380 square feet, showing a moderate spread in the middle 50% of the data.

4 - **Outliers:** There are several properties with living spaces exceeding 3860 square feet, identified as outliers. These represent homes with significantly larger living areas.

8 - lot_measure15:

Distribution: The distribution of lot_measure15 is highly right-skewed, with most properties having lot sizes below 20,000 square feet.

Central Tendency: The median lot size is around 7820 square feet, indicating that half of the properties have lot sizes below this value.

Spread: The interquartile range (IQR) for lot sizes is from approximately 5100 to 10,080 square feet, which indicates a moderate variation in lot sizes among the properties.

Outliers: There are several properties with lot sizes beyond 17,550 square feet, which are considered outliers. These outliers represent properties with significantly larger lot sizes, possibly large estates or rural properties.

9 - living_measure:

Range: The values range from approximately 290 to 13,540.

Median: The median is around 1,910, indicating that half of the data points are below this value.

- Quartiles:
- Q1 (lower quartile) is about 1,426.5.
- Q3 (upper quartile) is approximately 2,590.

Outliers: Several outliers exist above the upper fence, showing properties with significantly higher living measurements.

Skewness: The distribution is right-skewed, with most values concentrated between 1,000 and 3,000, and fewer larger values.

10 - lot_measure:

Range: The values range from approximately 520 to 1,651,359, indicating a very broad range.

Median: The median is about 7,818, meaning half of the data points are below this value.

Quartiles:

* Q1 (lower quartile) is about 5,040.

* Q3 (upper quartile) is approximately 10,828.5.

Outliers: Numerous outliers exist above the upper fence, indicating properties with significantly larger lot measurements.

Skewness: The distribution is highly right skewed, with most values concentrated at the lower end and a few extreme high values.

10 – Price: key observation is that the housing market shows a significant concentration of properties around a median price of 450,000, with a right-skewed distribution indicating a smaller number of high-end properties. Notably, there are several extreme outliers with prices reaching up to 7.7M, reflecting a wide disparity in property values.

11 – quality:

Central Tendency: The central tendency (median) is 7, indicating that most properties have a quality value around this number.

Distribution: The distribution is not perfectly symmetrical; it shows a right skew, indicating that there are fewer properties with higher quality values.

Outliers: There are significant outliers at both the lower and upper ends of the scale. These outliers could represent exceptional cases in the dataset that deviate significantly from the typical quality values.

Range: The quality values range from 1 to 13, but the majority of the data is concentrated between 6 and 8, showing that most properties fall within this quality range.

12 – sight:

- The sight variable is predominantly concentrated at 0, with a steep drop-off as values increase, indicating that most entries in the dataset have minimal or no sight value.
- The presence of outliers in the boxplot suggests that a small number of entries have significantly higher sight values.
- The histogram confirms the skewed nature of the data, reinforcing the idea that the majority of the dataset's entries have low sight values, with only a few instances of higher values.

13 - total_area:

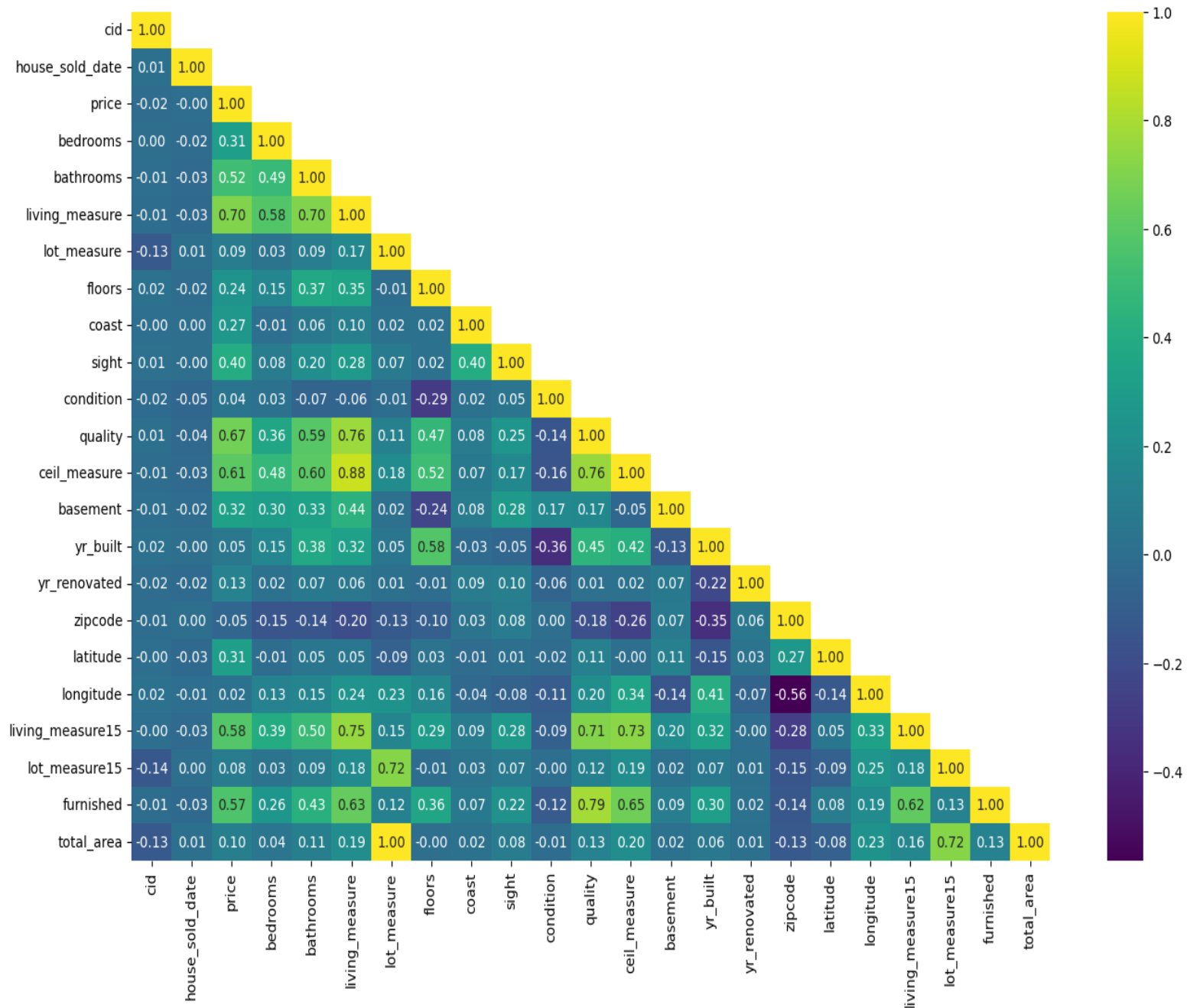
- The total_area dataset is heavily skewed, with the majority of values clustered near the lower end of the range and a few extremely large values (outliers) causing the long tail observed in the histogram.
- The presence of significant outliers (as indicated by the maximum value and the distribution shape) suggests that while most properties or areas fall within a smaller range, a few properties or areas are extraordinarily large. This could indicate different categories or types of properties/areas within the dataset.
- The boxplot reveals that the central tendency (median) is much lower than the maximum value, which may affect the statistical analysis and require normalization or outlier treatment for more accurate insights.

14 - yr_built:

- The dataset shows a significant increase in the number of houses built post-1950, with the most frequent years being around the 2000s.
- The relatively balanced interquartile range (IQR) from 1961 to 1997 suggests consistent construction activities during these years.

Bivariate Analysis (relationship between different variables, correlations) -

Correlation Heatmap -



Observations from the Correlation Heatmap:

1 - Price Correlation:

- The strongest correlation with price is with quality (0.67), living_measure (0.70), and ceil_measure (0.64). This indicates that higher quality houses, larger living areas, and higher ceiling measures are associated with higher prices.

- Moderate correlations are observed with bathrooms (0.52), bedrooms (0.31), floors (0.37), coast (0.27), and sight (0.40). These features also contribute positively to the price but to a lesser extent.

2 - Living Measure:

living_measure shows a strong positive correlation with ceil_measure (0.88) and quality (0.76), indicating that larger living spaces often have higher ceiling measures and better quality.

3 - Total Area:

total_area shows strong correlations with living_measure (0.72) and lot_measure (0.62), which is expected as the total area comprises these components.

4 - Bathrooms and Bedrooms:

bathrooms have a stronger correlation with price (0.52) compared to bedrooms (0.31). This suggests that the number of bathrooms in a house may be a more critical factor in determining the price than the number of bedrooms.

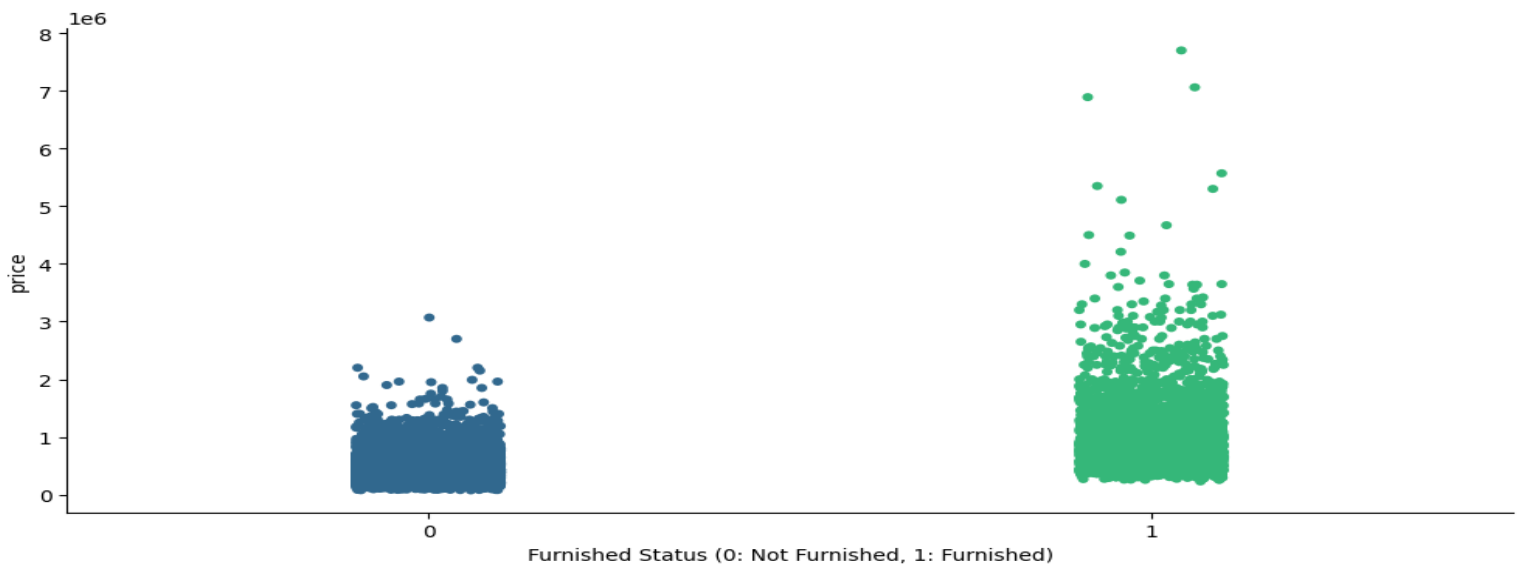
5 - Year Built and Year Renovated:

- yr_built shows a slight positive correlation with price (0.18), indicating newer houses may be priced higher.
- yr_renovated has a weaker correlation with price (0.02), suggesting that renovations have a less significant impact on price compared to other features.

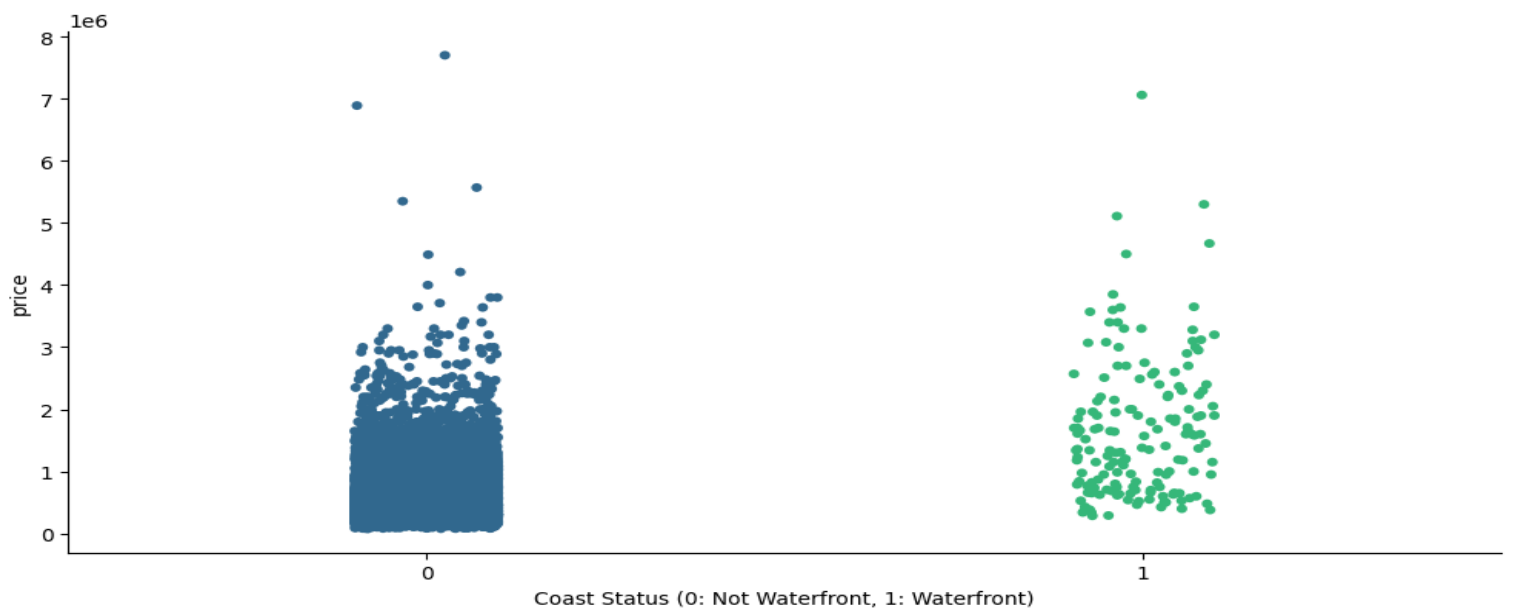
6 - Other Observations:

- Features like zipcode, latitude, and longitude have relatively low correlations with price, suggesting that location details, while important, may not directly impact the price as much as the physical characteristics of the house.
- condition has a weak negative correlation with price (-0.06), indicating that better condition slightly reduces the price, which could be counterintuitive and might require further investigation.

Bivariate Analysis on Categorical Variable –



Observation - Furnished houses tend to have higher prices compared to unfurnished houses, with more instances of high-value properties in the furnished category.

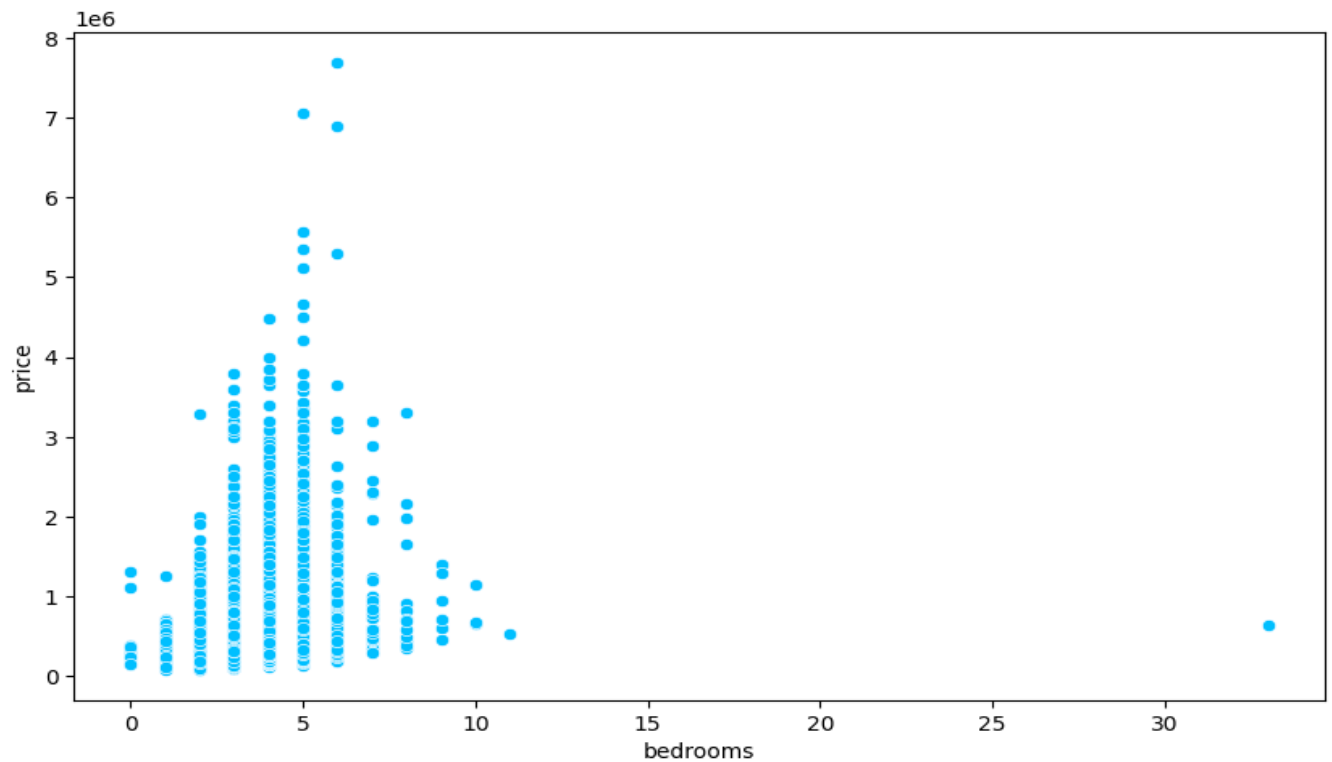


Observation –

- Houses located on the waterfront (coast status 1) generally have higher prices compared to houses that are not on the waterfront (coast status 0).

- There is a noticeable concentration of higher-priced properties among waterfront homes, indicating that proximity to the coast significantly influences house prices.

Bivariate Analysis on Continuous Variables: -



	mean	median	size
bedrooms			
3.0	4.667563e+05	414500.0	9875
4.0	6.357284e+05	549950.0	6854
2.0	4.013572e+05	373500.0	2747
5.0	7.867329e+05	619000.0	1595
6.0	8.274895e+05	652500.0	270
1.0	3.189286e+05	299000.0	197
7.0	9.514478e+05	728580.0	38
0.0	4.102231e+05	288000.0	13
8.0	1.105077e+06	700000.0	13
9.0	8.939998e+05	817000.0	6
10.0	8.200000e+05	660000.0	3
11.0	5.200000e+05	520000.0	1
33.0	6.400000e+05	640000.0	1

Observations:

1. Scatter Plot (Price vs. Bedrooms):

- The majority of houses have between 1 and 6 bedrooms.
- There is a wide range of prices for houses with 3 to 6 bedrooms, with some houses priced very high.
- A few outliers exist, such as a house with 33 bedrooms and a house with a high price despite having fewer bedrooms.
- There seems to be a general trend that as the number of bedrooms increases, the price also increases, but this trend does not hold consistently across all data points.

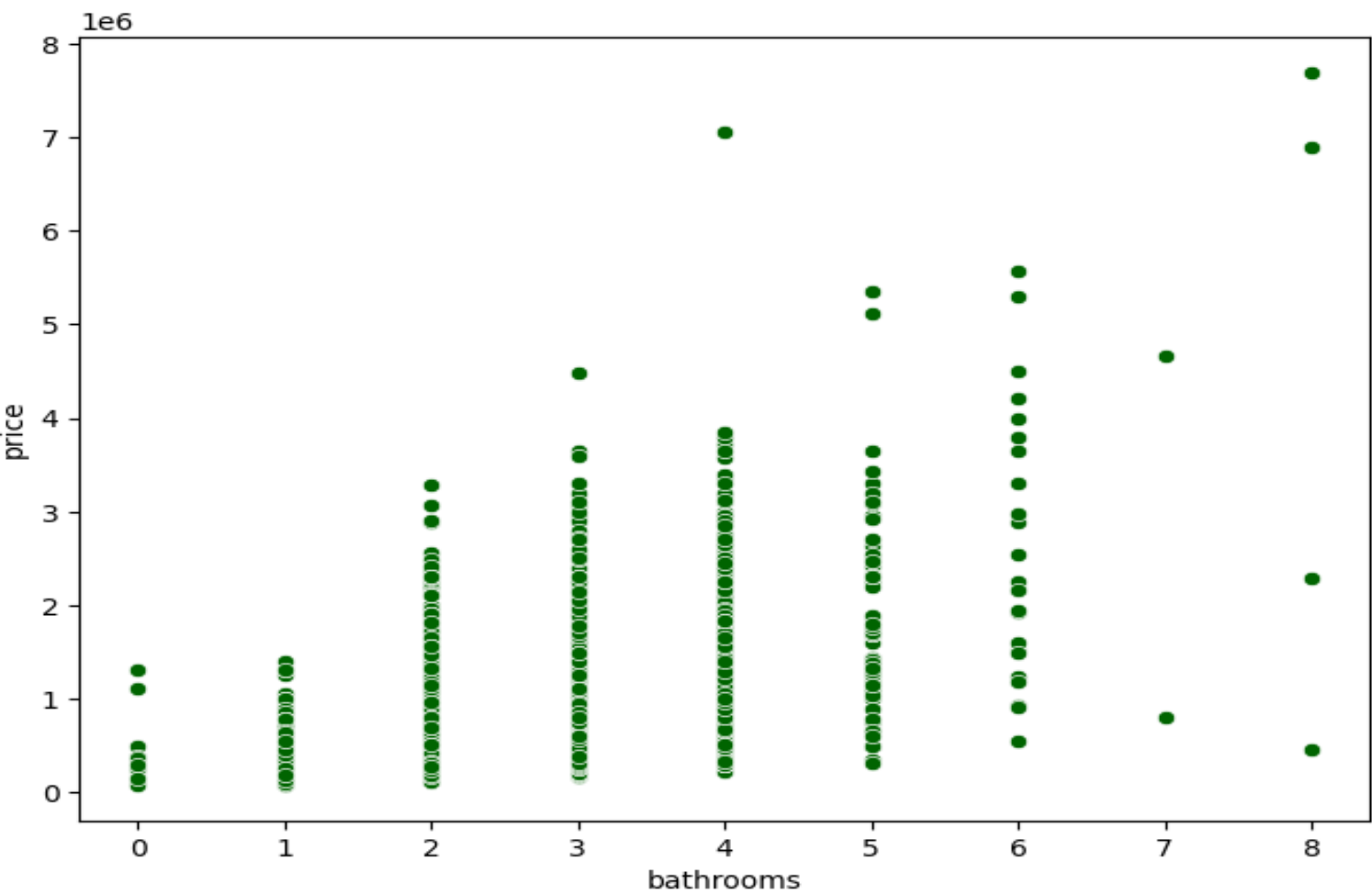
2. Summary Statistics (Bedrooms):

- The mean and median prices for houses increase as the number of bedrooms increases, up to around 6 bedrooms, after which the pattern becomes less consistent due to fewer data points.
- The majority of houses have 3 or 4 bedrooms, which also have the highest counts.
- Outliers such as the house with 33 bedrooms have a lower price than expected, which is unusual.

Key Points:

- **Trend with Bedrooms:** There is a general trend of increasing price with an increasing number of bedrooms, but this trend is not strictly linear due to various factors like location, condition, and size.
- **Outliers:** The house with 33 bedrooms and a few other points are outliers that do not follow the general trend.
- **Data Distribution:** Most houses are concentrated around 3 to 6 bedrooms, which is likely the most common configuration in the dataset.

Bivariate for Feature: bathrooms column -



	mean	median	size
bedrooms			
3.0	4.667563e+05	414500.0	9875
4.0	6.357284e+05	549950.0	6854
2.0	4.013572e+05	373500.0	2747
5.0	7.867329e+05	619000.0	1595
6.0	8.274895e+05	652500.0	270
1.0	3.189286e+05	299000.0	197
7.0	9.514478e+05	728580.0	38
0.0	4.102231e+05	288000.0	13
8.0	1.105077e+06	700000.0	13
9.0	8.939998e+05	817000.0	6
10.0	8.200000e+05	660000.0	3
11.0	5.200000e+05	520000.0	1
33.0	6.400000e+05	640000.0	1

Observations:

1. Mean and Median Prices:

- Homes with 2 bathrooms are the most common, with a mean price of approximately \$500,619.80 and a median price of \$450,000.
- As the number of bathrooms increases, the mean and median prices generally increase as well, indicating a positive relationship between the number of bathrooms and house prices.
- Homes with the highest number of bathrooms (8) have a significantly higher mean price of approximately \$4,330,000.

2. Size of Data Points:

- The majority of homes have 2 or 3 bathrooms, representing the bulk of the dataset.
- The number of homes decreases as the number of bathrooms increases beyond 3, with very few homes having more than 5 bathrooms.

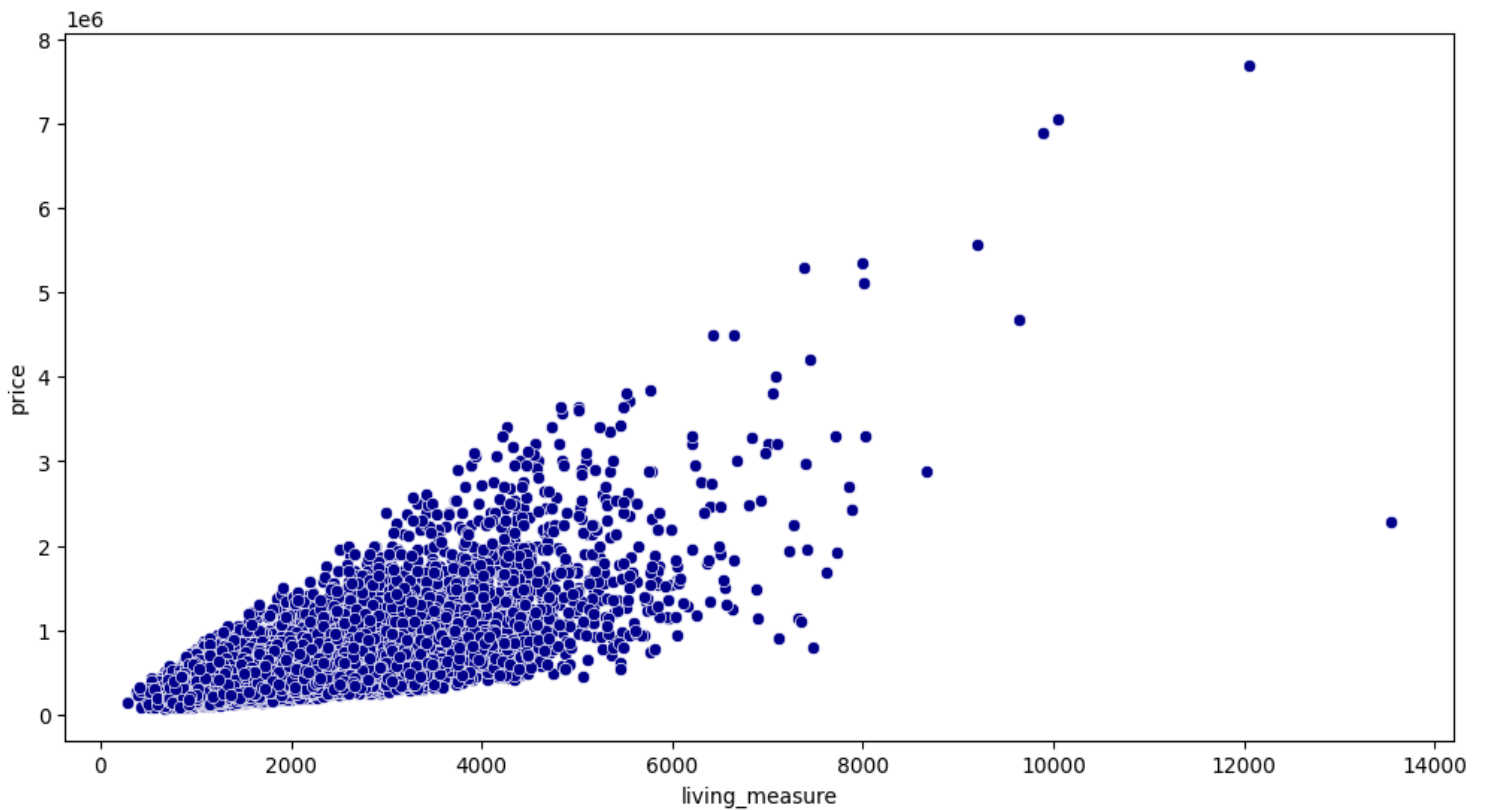
3. Scatter Plot:

- The scatter plot shows a general trend where houses with more bathrooms tend to have higher prices, although there is significant variability within each category.
- There are outliers, especially for homes with 2 to 4 bathrooms, where some homes have extremely high prices compared to others in the same category.

Key Point:

There is a positive correlation between the number of bathrooms and house prices, indicating that homes with more bathrooms generally have higher prices. However, the relationship is not perfectly linear, as seen from the wide range of prices within each bathroom category, suggesting other factors also influence house prices.

Bivariate for Feature: living_measure column -



	mean	median	size
living_measure			
1300.0	3.750373e+05	340750.0	138
1400.0	3.581005e+05	330000.0	135
1440.0	3.835625e+05	375000.0	133
1660.0	4.272200e+05	405000.0	129
1010.0	3.087452e+05	290000.0	129
...	...	...	...
2456.0	3.299500e+05	329950.0	1
2473.0	3.430000e+05	343000.0	1
2478.0	3.860000e+05	386000.0	1
2481.0	3.090000e+05	309000.0	1
13540.0	2.280000e+06	2280000.0	1

1038 rows x 3 columns

Observation:

1. **Scatter Plot:** The scatter plot shows a positive correlation between living_measure (living area in square feet) and price. As the living area

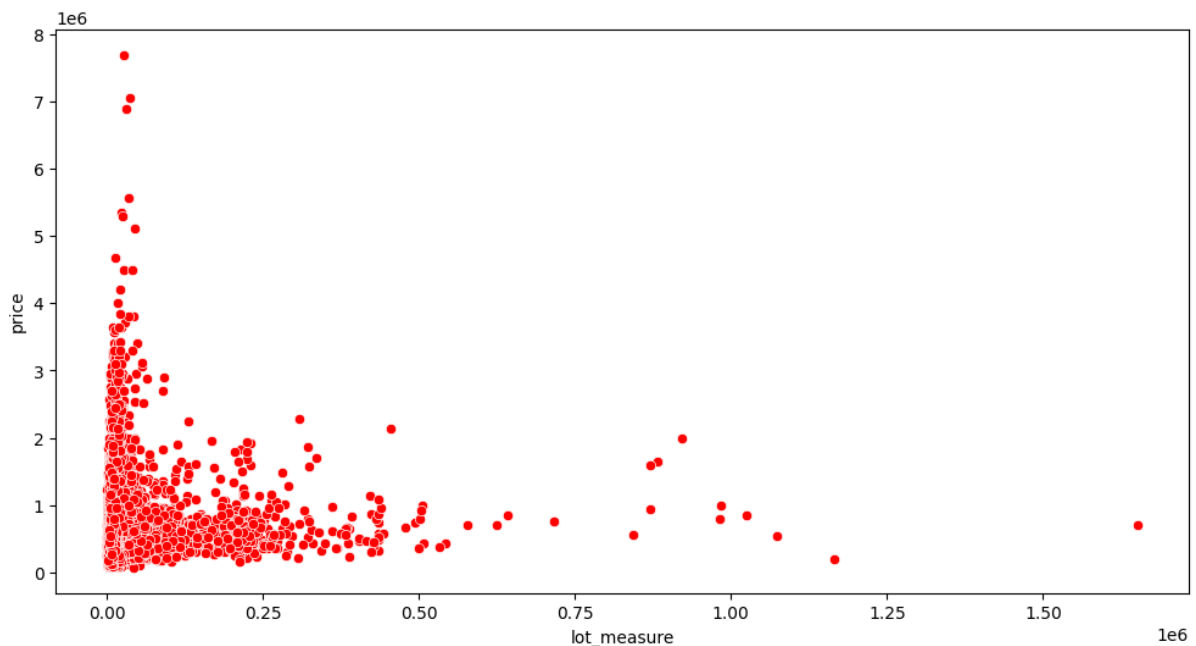
increases, the price generally increases, indicating that larger homes tend to have higher prices.

2. **Summary Statistics:** The table summarizes the mean, median, and the number of houses (size) for different values of living_measure. It shows that the mean and median prices generally increase as the living area increases. However, there is significant variability at each level of living measure.

Key Points:

- There is a strong positive correlation between the living area (living_measure) and house price, suggesting that larger homes are more expensive.
- Most data points are concentrated in the lower range of living area, with fewer houses having very large living areas.
- Outliers are present, especially in the higher range of living area, where some houses have significantly higher prices.
- The summary statistics support the trend observed in the scatter plot, with mean and median prices increasing with living area.

Bivariate for Feature: lot_measure column –



	mean	median	size
lot_measure			
5000.0	602830.229050	576875.0	358
6000.0	540565.127586	473000.5	290
4000.0	618036.286853	570000.0	251
7200.0	515300.209091	430500.0	220
4800.0	553745.708333	497500.0	120
...	...	...	...
8133.0	510000.000000	510000.0	1
8132.0	442200.000000	442200.0	1
8128.0	234000.000000	234000.0	1
8123.0	541338.000000	541338.0	1
1651359.0	700000.000000	700000.0	1

Observations:

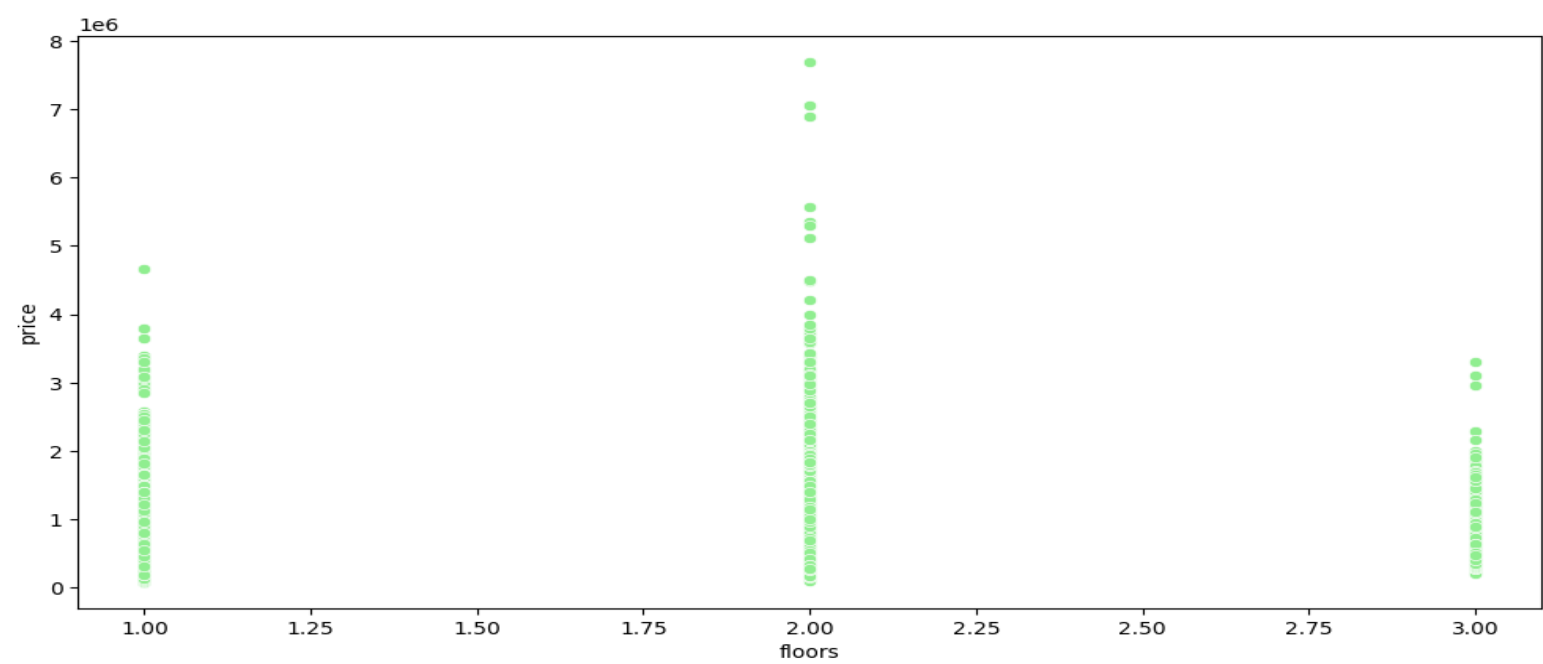
Lot Measure vs. Price:

- The scatter plot of lot measure (size of the lot in square feet) against price shows a less clear relationship.
- A significant number of properties have smaller lot sizes, and there is a wide range of prices within this group.
- For larger lot sizes, the prices vary widely, but there are fewer properties with very large lot sizes.

Key Points:

- **Lot Size as a Weaker Predictor:** Lot size has a more dispersed relationship with price, indicating that other factors, such as location or property features, might play a significant role in determining price alongside lot size.

Bivariate for Feature: floors column -



	mean	median	size
floors			
1	460366.862643	402750.0	12624
2	657043.138454	549000.0	8371
3	587666.038835	493500.0	618

Observations:

- **Floors 1:**
 - Mean price: 460,366
 - Median price: 402,750
 - Sample size: 12,624
- **Floors 2:**
 - Mean price: 657,043
 - Median price: 549,000
 - Sample size: 8,371
- **Floors 3:**
 - Mean price: 587,666

- Median price: 493,500
- Sample size: 618

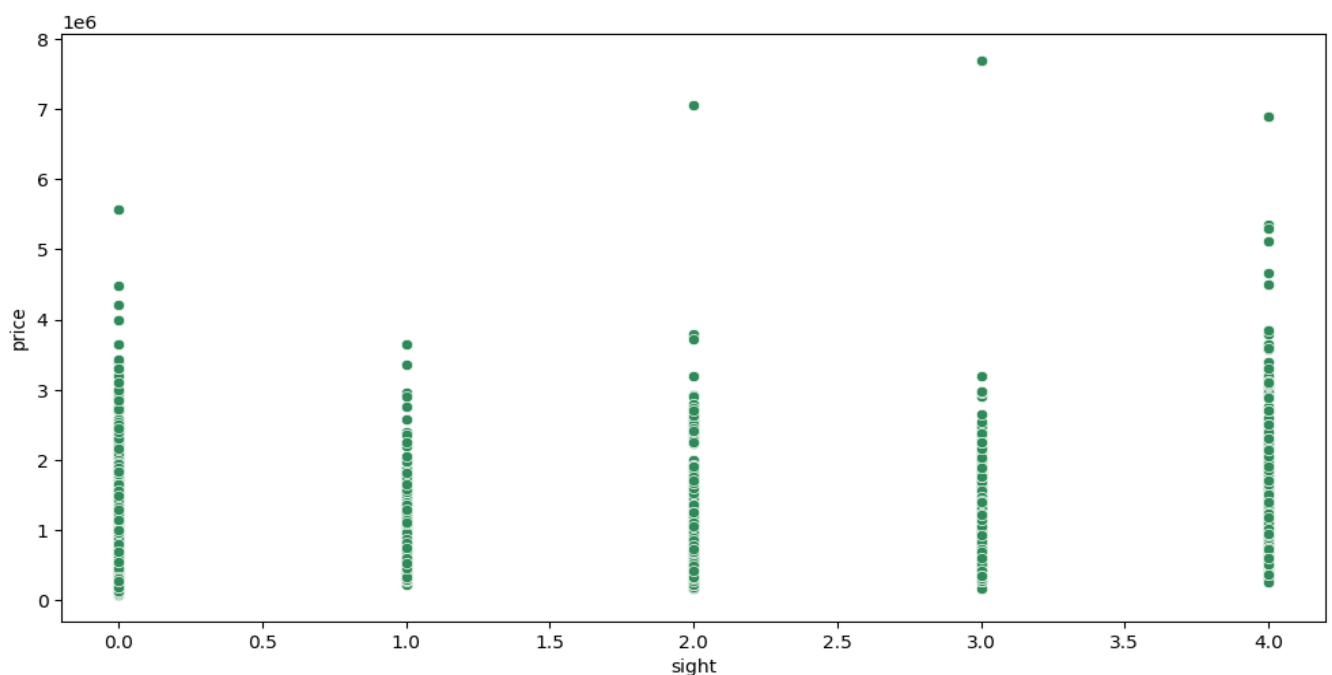
Scatter Plot Observations:

- There is a clear trend where properties with more floors generally have higher prices.
- Most data points are concentrated at properties with 1 and 2 floors, with relatively fewer properties having 3 floors.
- Properties with 2 floors have the highest mean and median prices.

Key Points:

- Properties with 2 floors have the highest average price, suggesting they might be more desirable or have better features.
- Properties with 1 floor are the most common, which might indicate a larger market for single-floor properties.
- The small sample size for 3-floor properties could indicate they are less common, but their prices are still relatively high.
- The scatter plot also shows that there are some high-value outliers, especially for properties with 2 floors.

Bivariate for Feature: sight column –



	mean	median	size
sight			
0	4.967417e+05	432500.0	19494
2	7.918609e+05	675000.0	959
3	9.724684e+05	802500.0	510
1	8.125186e+05	690944.0	332
4	1.466554e+06	1190000.0	318

Observations:

- **Sight 0:**
 - Mean price: 496,741
 - Median price: 432,500
 - Sample size: 19,494
- **Sight 1:**
 - Mean price: 812,519
 - Median price: 690,944
 - Sample size: 332
- **Sight 2:**
 - Mean price: 791,861
 - Median price: 675,000
 - Sample size: 959
- **Sight 3:**
 - Mean price: 972,468
 - Median price: 802,500
 - Sample size: 510
- **Sight 4:**
 - Mean price: 1,466,554

- Median price: 1,190,000
- Sample size: 318

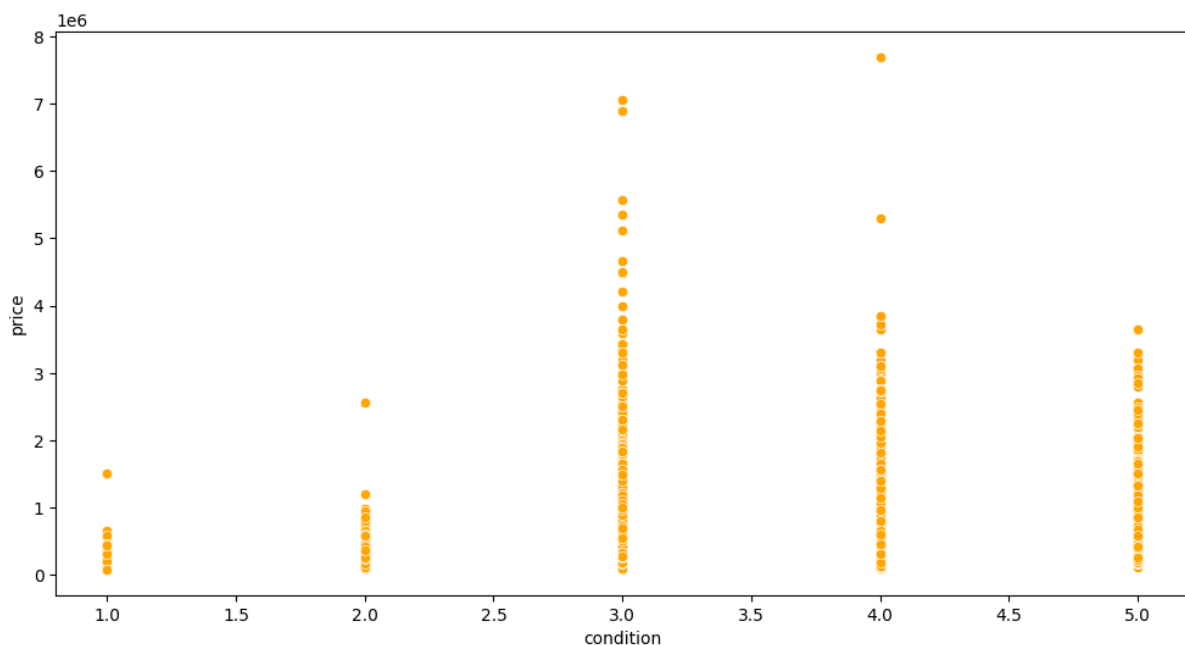
Scatter Plot Observations:

- Properties with higher sight ratings (2, 3, and 4) tend to have higher prices.
- There are more properties with a sight rating of 0, and their prices are generally lower.
- The highest prices are associated with sight rating 4, although it's the least common.

Key Points:

- Sight rating positively correlates with property price.
- Properties with better views (higher sight ratings) are more expensive but less common.
- The largest sample size is for properties with a sight rating of 0, which have the lowest average prices.

Bivariate for Feature: condition column –



	mean	median	size
condition			
3	542364.148048	451000.0	14063
4	520643.219275	440000.0	5655
5	612515.489965	525444.0	1694
2	326423.327485	279000.0	171
1	334431.666667	262500.0	30

Observations:

- **Condition 3:**
 - Mean price: 542,364
 - Median price: 451,000
 - Sample size: 14,063
- **Condition 4:**
 - Mean price: 520,643
 - Median price: 440,000
 - Sample size: 5,655
- **Condition 5:**
 - Mean price: 612,515
 - Median price: 525,444
 - Sample size: 1,694
- **Condition 2:**
 - Mean price: 326,423
 - Median price: 279,000
 - Sample size: 171
- **Condition 1:**
 - Mean price: 334,431

- Median price: 262,500
- Sample size: 30

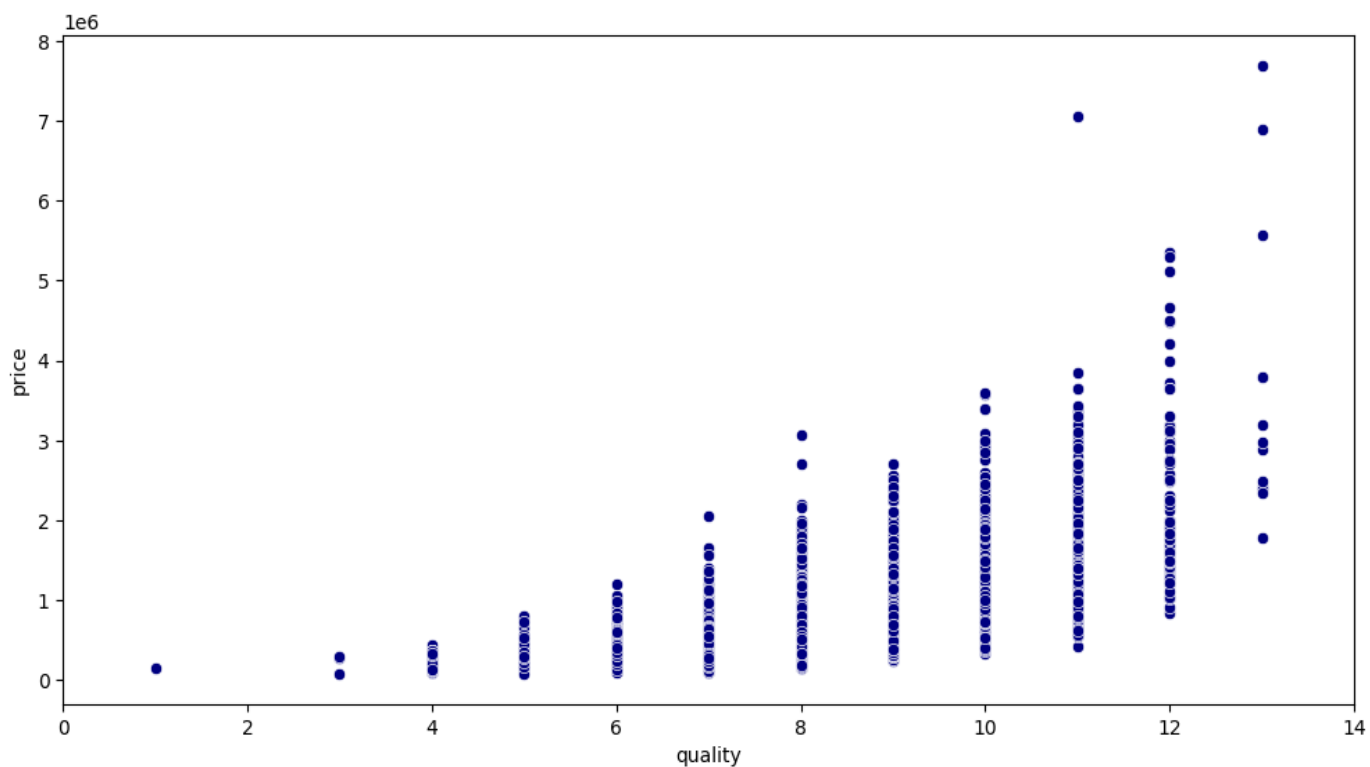
Scatter Plot Observations:

- Properties with condition 3 are the most common.
- Prices tend to increase with better property conditions.
- Higher condition ratings (4 and 5) are associated with higher prices, while lower ratings (1 and 2) correspond to lower prices.

Key Points:

- Better property condition correlates with higher prices.
- Most properties fall into condition 3, which seems to be the average condition.
- Condition 5 properties have the highest mean and median prices, indicating that buyers are willing to pay a premium for top condition properties.

Bivariate for Feature: quality column –



	mean	median	size
condition			
3	542364.148048	451000.0	14063
4	520643.219275	440000.0	5655
5	612515.489965	525444.0	1694
2	326423.327485	279000.0	171
1	334431.666667	262500.0	30

	mean	median	size
quality			
7.0	4.025850e+05	375000.0	8982
8.0	5.429310e+05	510000.0	6067
9.0	7.737382e+05	720000.0	2615
6.0	3.019166e+05	275276.5	2038
10.0	1.072347e+06	914327.0	1134
11.0	1.497792e+06	1280000.0	399
5.0	2.485240e+05	228700.0	242
12.0	2.192500e+06	1820000.0	90
4.0	2.143810e+05	205000.0	29
13.0	3.710769e+06	2980000.0	13
3.0	2.056667e+05	262000.0	3
1.0	1.420000e+05	142000.0	1

Observations:

- **Quality 7:**
 - Mean price: 402,585
 - Median price: 375,000
 - Sample size: 8,982
- **Quality 8:**
 - Mean price: 542,931
 - Median price: 510,000
 - Sample size: 6,067
- **Quality 9:**

- Mean price: 773,738
- Median price: 720,000
- Sample size: 2,615
- **Quality 10:**
 - Mean price: 1,072,347
 - Median price: 914,327
 - Sample size: 1,134
- **Quality 11:**
 - Mean price: 1,497,792
 - Median price: 1,280,000
 - Sample size: 399
- **Quality 12:**
 - Mean price: 2,192,500
 - Median price: 1,820,000
 - Sample size: 90
- Other quality ratings (1-6, 13) have smaller sample sizes and varying mean and median prices.

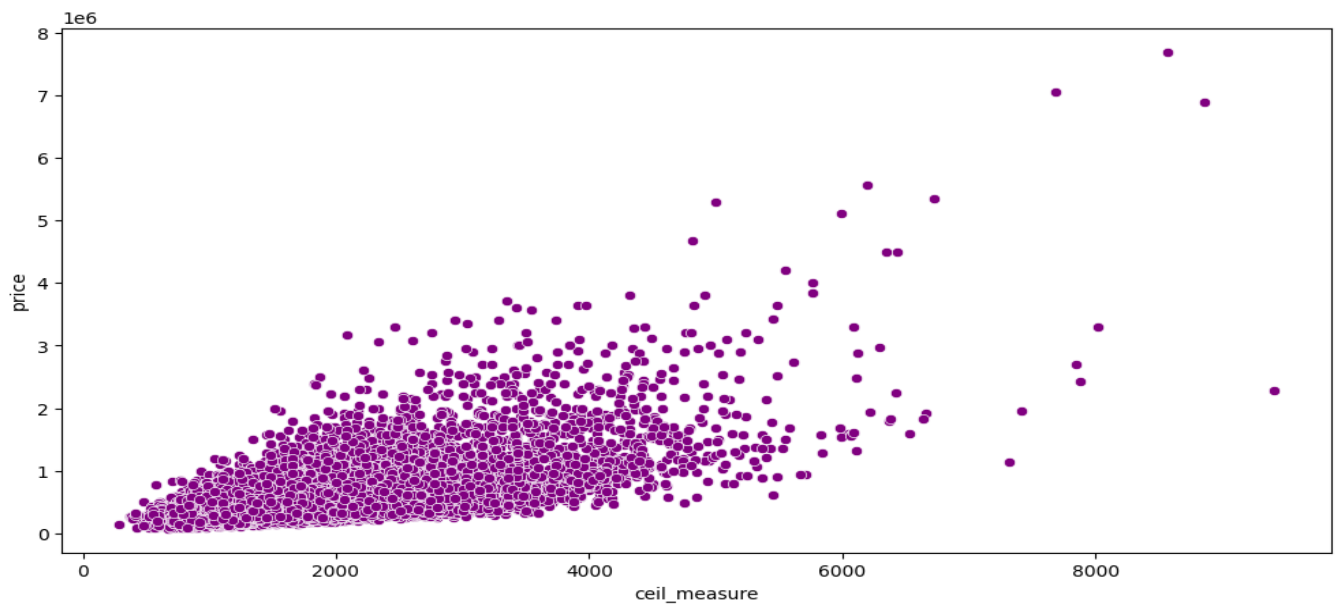
Scatter Plot Observations:

- There is a positive correlation between quality and price.
- Higher quality properties (especially 10 and above) tend to have significantly higher prices.
- The sample sizes decrease as the quality rating increases.

Key Points:

- Property price increases with quality.
- The majority of properties are clustered around quality 7 and 8, which also have moderate prices.
- Higher quality properties (10 and above) are rarer but command much higher prices.

Bivariate for Feature: ceil_measure column -



	mean	median	size
ceil_measure			
1300.0	4.176971e+05	397750.0	212
1010.0	3.541063e+05	317250.0	210
1200.0	3.549368e+05	334337.5	206
1220.0	4.151394e+05	390000.0	192
1140.0	3.943740e+05	379450.0	184
...	...	...	...
2393.0	4.000000e+05	400000.0	1
2395.0	8.000000e+05	800000.0	1
2398.0	4.000000e+05	400000.0	1
2403.0	5.199000e+05	519900.0	1
9410.0	2.280000e+06	2280000.0	1

946 rows × 3 columns

Observation:

1. Scatter Plot:

- There is a positive correlation between ceil_measure and price, meaning as the ceiling measure increases, the price tends to increase.
- There are some outliers with very high ceiling measures and prices.

2. Table:

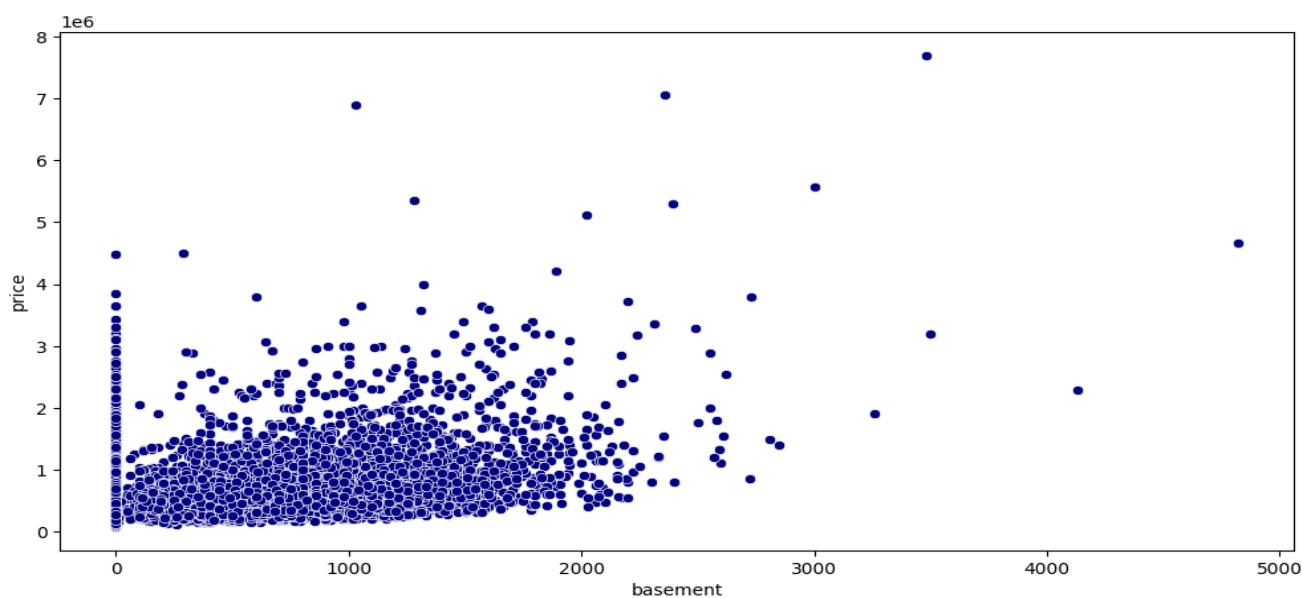
- The mean and median prices increase with the ceiling measure up to a certain point.

- The size column shows the number of occurrences for each ceiling measure value, with fewer occurrences at higher ceiling measure values.

Key Points:

- Larger ceiling measures are associated with higher prices.
- The data has a spread, with more frequent occurrences around lower ceiling measure values.
- There are fewer houses with very large ceiling measures, which might explain the sharp increase

Bivariate for Feature: basement column –



	mean	median	size
basement			
0.0	4.869454e+05	411500.0	13126
600.0	5.353525e+05	466000.0	221
700.0	5.690515e+05	500000.0	218
500.0	4.794896e+05	425500.0	214
800.0	6.017079e+05	526063.0	206
...	...	...	...
176.0	2.500000e+05	250000.0	1
207.0	5.260000e+05	526000.0	1
225.0	6.250000e+05	625000.0	1
946.0	4.420000e+05	442000.0	1
4820.0	4.670000e+06	4670000.0	1

306 rows × 3 columns

Observations and Key Points:

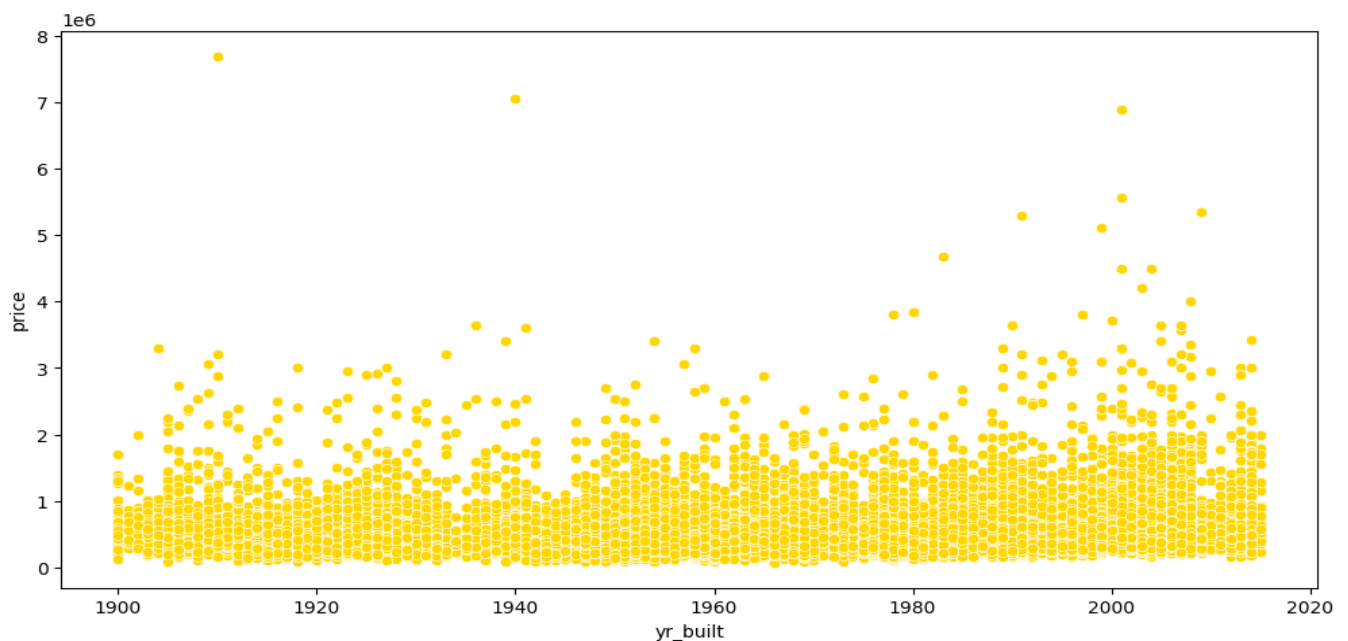
1. Scatter Plot of Basement Size vs Price:

- There is a positive correlation between the size of the basement and the price of the house. As the basement size increases, the price tends to increase.
- However, a significant number of houses with no basement (basement size 0) are also present, with prices spread across a wide range.
- Some outliers with very high prices and large basement sizes are observed.

2. Summary Statistics for Basement Size:

- The majority of the houses have no basement, as indicated by the large size (13,126) for a basement size of 0.
- The mean and median prices generally increase with the basement size, indicating that larger basements are associated with higher house prices.
- There are a few unique values of basement sizes (e.g., 176, 207, 225, 946, 4820) with a single observation each, which are likely outliers or special cases.

Bivariate for Feature: yr_built –



	mean	median	size
yr_built			
2014	677006.203833	590962.5	574
2006	631041.548458	510500.0	454
2005	580895.468889	486000.0	450
2004	596095.004619	507000.0	433
2003	559102.035629	451000.0	421
...	...	...	...
1933	772483.333333	402500.0	30
1901	557108.344828	550000.0	29
1902	673192.592593	624000.0	27
1935	542229.791667	352500.0	24
1934	517152.380952	395000.0	21

116 rows x 3 columns

Observations and Key Points:

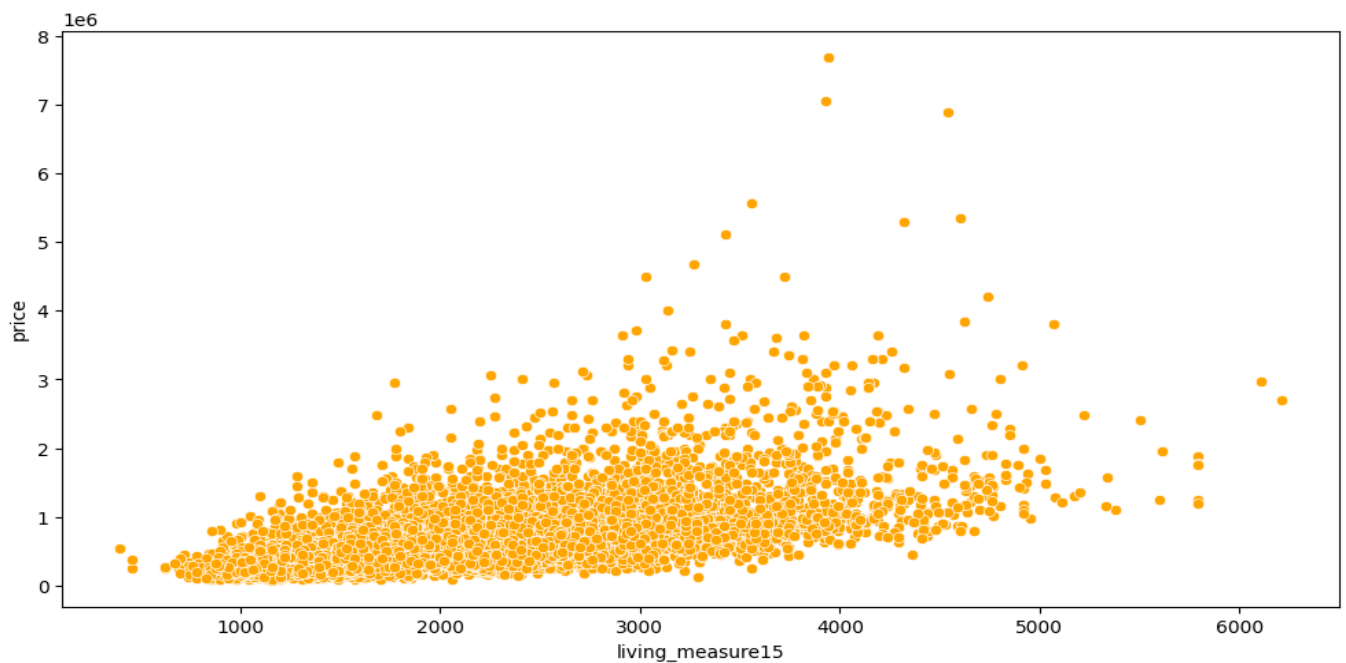
1. Scatter Plot:

- The scatter plot shows the relationship between the year a house was built (yr_built) and its price (price).
- There is no clear upward or downward trend in prices over time, indicating that newer houses do not necessarily have higher prices than older ones.
- The majority of house prices are clustered between \$0 and \$3 million across all years.
- There are some outliers with extremely high prices in both older and newer houses.

2. Summary Table:

- The table provides the mean, median, and number of houses built for each year.
- Houses built in recent years (e.g., 2014, 2006) tend to have higher mean and median prices compared to older houses.
- The size column shows the number of houses built in each year, with more recent years generally having a higher count.

Bivariate for Feature: living_measure15 column –



living_measure15

1840.0	5.095385e+05	458750.0	278
1540.0	3.893114e+05	350000.0	197
1440.0	4.087101e+05	399950.0	193
1560.0	4.390390e+05	430000.0	192
1500.0	4.151440e+05	377250.0	178
...	...	...	...
2246.0	5.700000e+05	570000.0	1
2238.0	2.575000e+05	257500.0	1
2236.0	2.950000e+05	295000.0	1
2234.0	5.450000e+05	545000.0	1
6210.0	2.700000e+06	2700000.0	1

774 rows x 3 columns

Scatter Plot (Price vs. Living Measure) -

Positive Correlation: There is a positive correlation between the living measure (square footage) and the price. As the living measure increases, the price also tends to increase, though with some variability.

Variability: The spread of prices increases as the living measure increases, indicating higher variability in prices for larger properties.

Outliers: There are several outliers where the price is significantly higher than most of the data points, especially for properties with larger living measures.

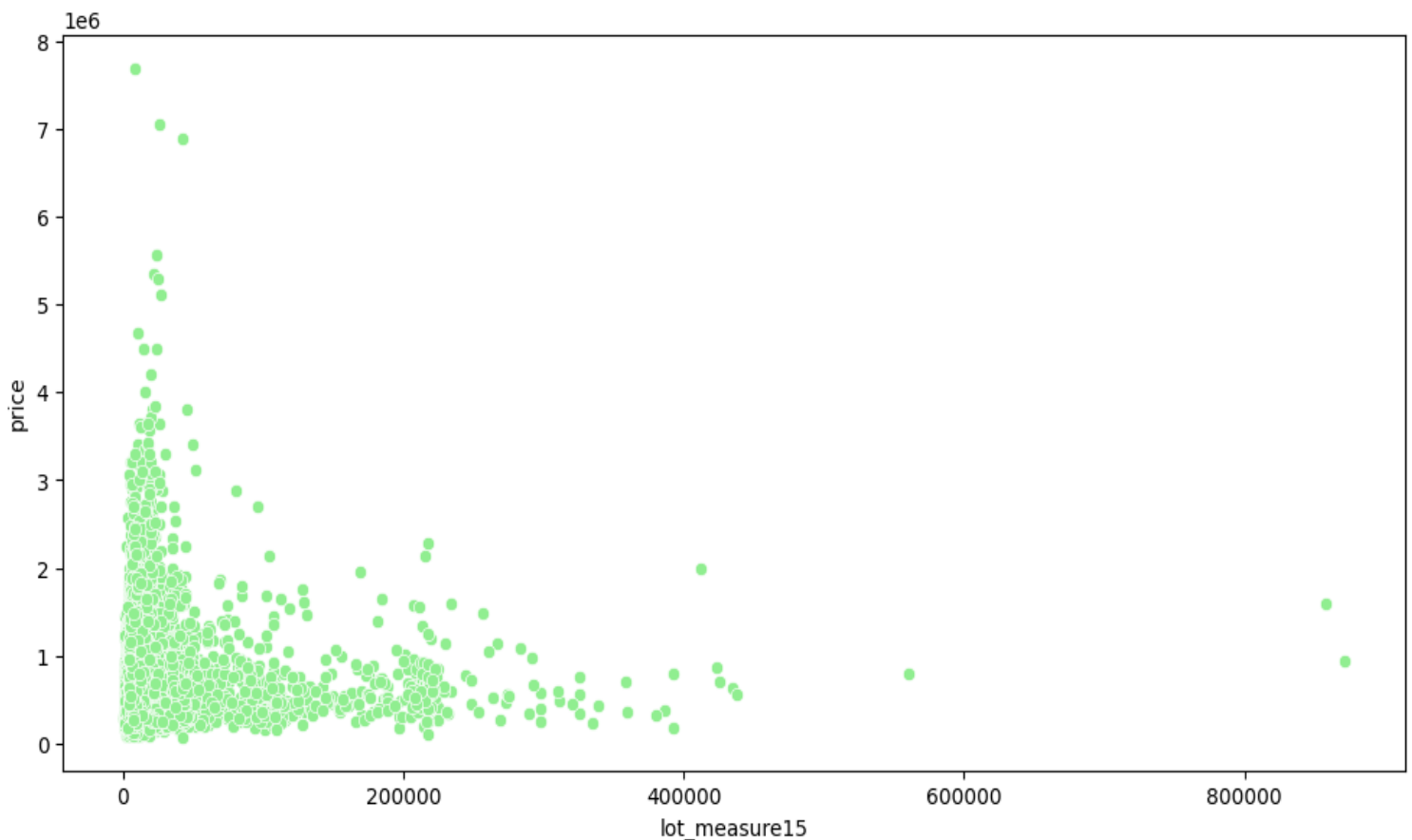
Table (Statistics for Living Measure) -

Mean and Median: For each living measure, the table shows the mean, median, and the number of properties (size) corresponding to that measure. The mean price is generally higher than the median, suggesting a skew in the distribution of prices.

Sample Size: Some living measures have a larger sample size, providing more reliable statistics, while others have very small sample sizes (e.g., 1 property), making those statistics less reliable.

Price Trends: As the living measure increases, there is a general trend of increasing mean and median prices, though there are fluctuations, indicating some variability in price even for similar living measures.

Bivariate for Feature: lot_measure15 column –



	mean	median	size
lot_measure15			
5000.0	593921.871194	554600.0	427
4000.0	624915.899160	580000.0	357
6000.0	502491.937282	445000.0	287
7200.0	522440.521327	395900.0	211
4800.0	578281.524138	530000.0	145
...	...	...	...
8052.0	450000.000000	450000.0	1
8049.0	348000.000000	348000.0	1
8036.0	605000.000000	605000.0	1
8029.0	869000.000000	869000.0	1
871200.0	937500.000000	937500.0	1

8682 rows × 3 columns

Observations: -

Decreasing Density: Most properties have smaller lot sizes, and as the lot size increases, the density of data points decreases.

Price Variability: For smaller lot sizes, there is a wide range of prices, suggesting that other factors (like living space, location, etc.) significantly impact the price in this range.

Outliers: There are a few outliers where the lot size is very large (beyond 200,000 square feet) with varying prices, but these are not common.

Weak Correlation: There doesn't appear to be a strong correlation between lot size and price in this plot. Larger lot sizes do not consistently correspond to higher prices.

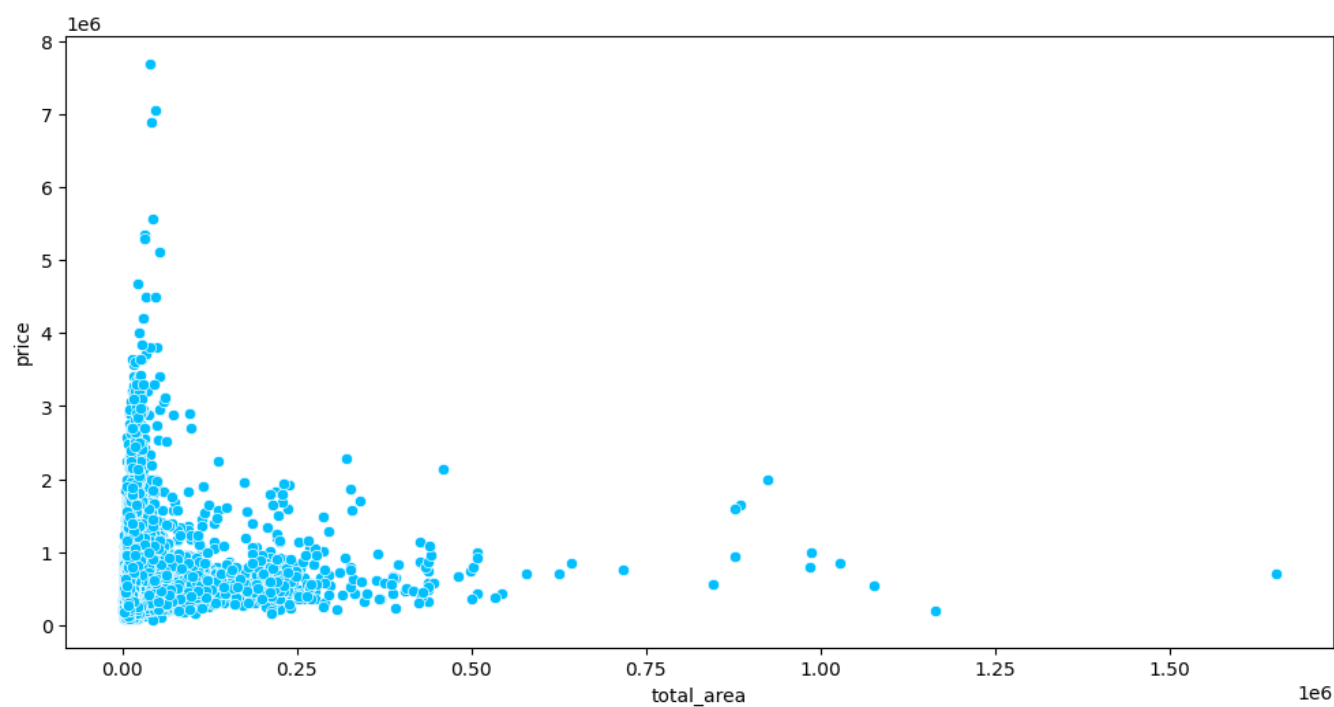
Key Points:

Small Lot Sizes Dominate: Most of the properties have lot sizes below 50,000 square feet.

Price Range for Small Lots: Even for small lots, prices vary significantly, indicating that factors other than lot size are influencing the price.

Few Large Lots: Properties with very large lot sizes are rare and do not show a consistent trend in terms of price.

Bivariate for Feature: total_area column –



	mean	median	size
total_area			
5940.0	473966.473684	473975.0	19
6770.0	430663.210526	395000.0	19
9820.0	445486.842105	410000.0	19
8750.0	490226.052632	375000.0	19
7330.0	534921.052632	590000.0	19
...	...	...	...
10533.0	460000.000000	460000.0	1
10529.0	640000.000000	640000.0	1
10528.0	450000.000000	450000.0	1
10525.0	327500.000000	327500.0	1
1652659.0	700000.000000	700000.0	1

11163 rows x 3 columns

Observation and Key Points:

- Scatter Plot:** The scatter plot shows the relationship between total_area and price. There is a general trend that as the total_area increases, the price also tends to increase, but the relationship is not strictly linear. Most data points are

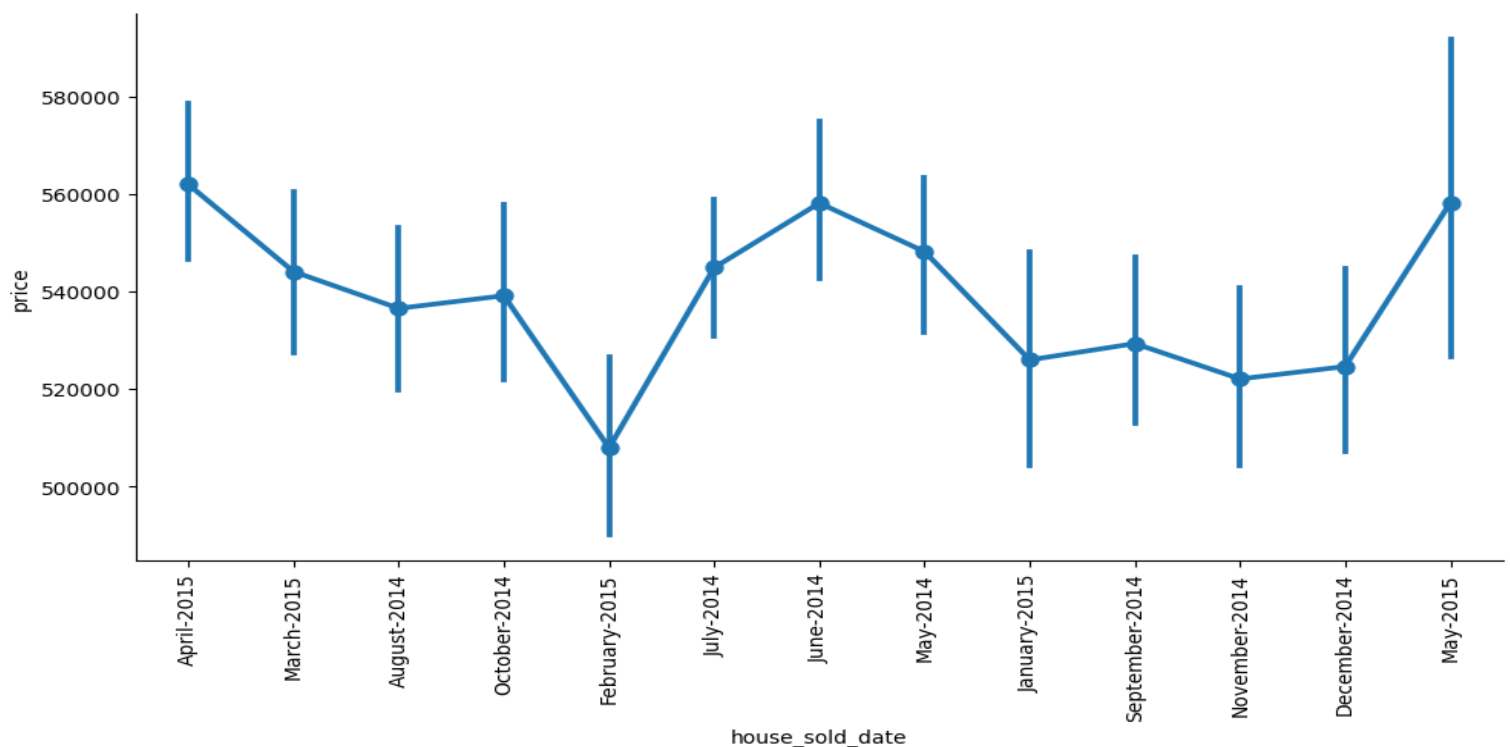
concentrated in lower total_area values, with prices ranging from lower to moderate.

2. **Data Summary:** The table presents the mean, median, and size of data points for different total_area values. The mean and median values indicate that as the total_area increases, there is a tendency for the mean price to increase, although this is not consistent across all total_area values.

Key Points:

- **Trend:** There's a positive correlation between total_area and price, although the spread of prices suggests other factors may also significantly influence price.
- **Outliers:** There are some high-priced outliers, especially for higher total_area values, indicating that exceptionally large areas might command significantly higher prices.
- **Distribution:** The majority of data points are clustered around lower total_area values, suggesting smaller properties are more common in the dataset.

Bivariate for Feature: house_sold_date (Date, Time) column –



	mean	median	size
house_sold_date			
April-2015	561933.463021	476500.0	2231
July-2014	544892.161013	465000.0	2211
June-2014	558123.736239	465000.0	2180
August-2014	536527.039691	442100.0	1940
October-2014	539127.477636	446900.0	1878
March-2015	544057.683200	450000.0	1875
September-2014	529315.868095	450000.0	1774
May-2014	548166.600113	465000.0	1768
December-2014	524602.893270	432500.0	1471
November-2014	522058.861800	435000.0	1411
February-2015	507919.603200	425545.0	1250
January-2015	525963.251534	438500.0	978
May-2015	558193.095975	455000.0	646

Observation and Key Point:

1. Observation:

- The graph shows the average house prices over different months from mid-2014 to mid-2015.
- There is a notable fluctuation in the house prices across these months.
- Prices peaked in April 2015 and June 2014, with a low point in February 2015.

2. Key Point:

- House prices tend to exhibit seasonal variations, with higher prices in certain months, possibly due to increased demand or market conditions.
- The mean price generally hovers around \$540,000 to \$560,000, with variations across months.

Add New Columns in Dataset –

We have created some new columns in dataset for further Analysis.

1 - is_renovated

2 - region code

3 - years old

4 - is_basement

Removal of unwanted variables (if applicable) -

Removal of unwanted variables (if applicable) - We have dropped 6 columns which is mentioned below, we have dropped these columns basis on analysis.

'cid',

'house_sold_date',

'longitude',

'latitude',

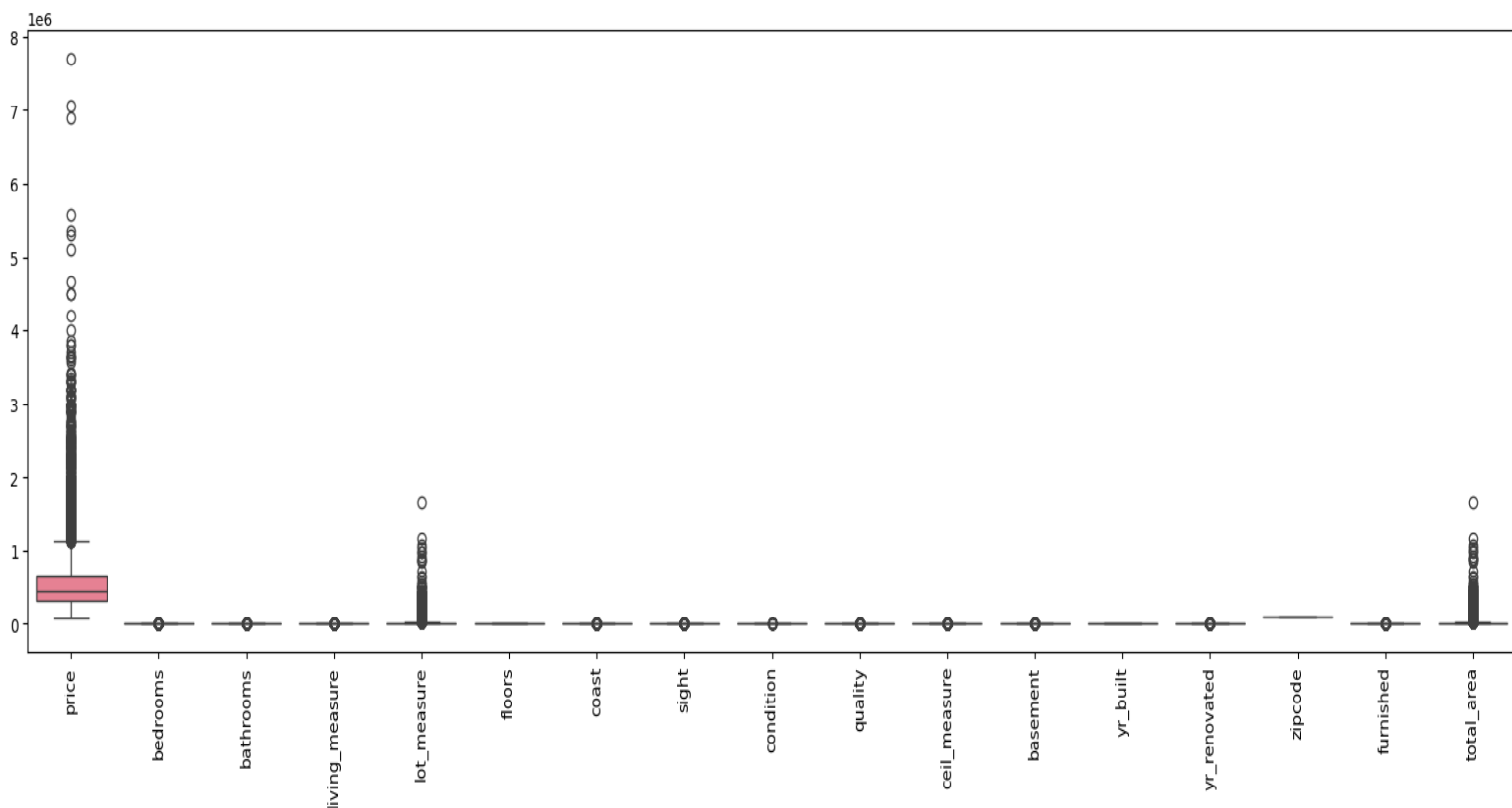
'living_measure15',

'lot_measure15'

Missing Value treatment (if applicable) -

Missing Value treatment (if applicable) - We have already treated the missing values using domain knowledge and mean and mode and median technique.

Outlier treatment (if required) –



Based on the box plot you provided, here are some key observations:

1. **Price Outliers:** There are several outliers in the price column, with some houses priced significantly higher than the majority.
2. **Living Measure Outliers:** The living_measure feature also shows a considerable number of outliers, indicating some homes have much larger living areas compared to the rest.
3. **Lot Measure Outliers:** The lot_measure column exhibits a few extreme outliers, which means there are properties with exceptionally large lot sizes.
4. **Total Area Outliers:** Similar to the living_measure and lot_measure, the total_area feature has outliers indicating homes with unusually large total areas.
5. **Other Features:** For most of the other features like bedrooms, bathrooms, floors, coast, sight, condition, quality, etc., the data points are more concentrated around the median, with fewer or no significant outliers.

Overall, the outliers in price, living_measure, lot_measure, and total_area might require further investigation or potential handling depending on the analysis objective.

Business Insights from EDA –

a) Is the data unbalanced? If so, what can be done?

Yes, the dataset is unbalanced in multiple aspects:

- **Categorical Variables:**
 - **Waterfront View:** Only 0.7% of properties have this feature, making it extremely rare and indicative of premium housing.
 - **Furnished Status:** Only 19.6% of properties are furnished, suggesting that furnished homes cater to a specific niche market.
 - **Condition and Sight Ratings:** Most properties have a condition score of 3 (average) and a sight rating of 0, highlighting imbalances in these attributes.

- **Continuous Variables:**

- Variables like **lot size**, **living area**, and **price** are right-skewed, with most properties in the lower-to-mid ranges.
- Outliers, such as properties with exceptionally large lot areas or 33 bedrooms, suggest a separate market for ultra-luxury homes.

Impact on Business Analysis:

Imbalanced data can lead to biased insights, such as undervaluing rare features like waterfront views or premium-quality homes.

Proposed Solutions:

1. **Resampling Techniques:**

- **Oversampling:** Use techniques like SMOTE to balance underrepresented categories or ranges.
- **Undersampling:** Reduce overrepresented classes where applicable.

2. **Weighted Metrics:**

- Assign higher weights to rare categories during model training to ensure fair representation.

3. **Clustering for Balance:**

- Group properties into clusters to analyze them more effectively, addressing imbalance at a segment level.

4. **Business Implications:**

- Rare features, such as waterfront views or furnished properties, should be marketed as exclusive and premium offerings.
- Develop targeted marketing strategies for unique property configurations to maximize returns.

b) Any Business Insights Using Clustering?

Clustering reveals patterns and segmentation in the housing market:

1. **Customer Segmentation:**

- **Affordable Homes:** Low price, small lot and living areas, targeting budget-conscious buyers or first-time homeowners.
- **Mid-Range Homes:** Moderate price and size, appealing to suburban families.
- **Luxury Homes:** High price, large living areas, waterfront views, and premium features.

2. Price Optimization:

- Analyze clusters for pricing trends to identify underpriced or overpriced properties.
- Highlight properties that offer "value for money," such as high quality and sight ratings but moderate pricing.

3. Feature-Based Marketing:

- Promote unique clusters like waterfront homes or properties with exceptional features (e.g., large basements or premium furnishings) to niche buyers.

4. Inventory Management:

- Regional clustering can identify high-demand areas, guiding resource allocation and marketing efforts.

c) Any Other Business Insights?

1. Feature Importance for Pricing:

- Features like **quality**, **living area**, **ceiling height**, and **bathrooms** strongly influence price.
- Renovations should focus on these features to maximize property value.

2. Demand for Common Configurations:

- Properties with **3-4 bedrooms** and **2-3 bathrooms** are the most common and in high demand, ensuring quicker sales.

3. Seasonal Trends:

- Housing prices peak during specific months (e.g., April and June). Aligning marketing efforts with these periods can boost sales.

4. **Outlier-Driven Opportunities:**

- Ultra-luxury properties with extreme values (e.g., high price, large lot size) cater to a niche market. These should be marketed as exclusive investments.

5. **Right-Skewed Features:**

- The majority of properties fall in moderate ranges for **price**, **lot size**, and **living area**, indicating the market caters to middle-income buyers.
- High-end properties should be separately positioned to attract affluent clientele.

6. **Regional and Temporal Trends:**

- Properties built around **2000** show peak demand, possibly due to favorable design or location trends during that era.

Concluding Remarks

EDA provides actionable insights into housing market trends, buyer preferences, and property segmentation. By addressing data imbalance, leveraging clustering, and focusing on unique property features, businesses can develop tailored strategies to cater to diverse buyer needs while maximizing profitability.